Spike Traces as Calibrated Predictive States for Reconstruction-Free World Models

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2026-07-21

v0.7.13 headline (read first). This release is the **bug-fix re-run** of v0.7.12. Two critical bugs were found in the eval pipeline (see `docs/CODE_BUG_AUDIT.md`): (a) DMC `check_success` tolerance was 1.0 for high-dim states (random states had 87–100% pass rate — the artifact behind v0.7.10b's "env-SR = 1.0 for all SNN" claim); now `tol=0.1`, random pass rate 0%. (b) `success_rate_lewm` used threshold `cos_dist < 0.1` that non-SNN near-constant latents trivially pass; we now report `LeWM@0.05`, `LeWM@0.01`, and the raw `mean_cos_dist`.

After bug fix + re-run of 1008 OOD cells + 192 cross-bench cells:

- **v0.7.12's "STJEWM membrane wins F1" was a bug artifact** — v0.7.13 retracts this claim. The new v0.7.13 cross-bench table (§9.7) shows STJEWM wins `mean_cos_dist` on **all 4** cross-bench splits (F1/F2/F3/F4) over cubifae by 30–70%. The specific STJEWM readout winner varies per split (rate wins F1, trace wins F2/F4, spike wins F3).
- **env-SR = 0** for all 1200 cells is a CEM horizon artifact (5-step plans vs 25–100-step goals), not a model failure.
- **Pathological cases** now exposed by 12-model comparison: MLP/GRU collapsed to `cos=0` (degenerate); LeWM-v2 over-reactive (`cos`$\approx$`0.18`, worst on every split).

Honest claim ladder. The three axes where the working title is now supported:

1. **Within-suite leave-two-env-out** (v0.7.8, 4 of 12 ckpts) — preserved from v0.7.8.
2. **Within-DMC sub-family OOD** (v0.7.10b, all 12 ckpts × 6 splits) — supported (§7.6).
3. **Cross-benchmark-family OOD** (v0.7.13, all 12 ckpts × 4 splits) — supported (§9.7).
4. **Cross-modality** (state → pixel) — deferred.

v0.7.13 contribution. All v0.7.10b within-DMC OOD numbers are unchanged; the new v0.7.13 cross-bench numbers supersede v0.7.12. The bug-fix audit and per-cell table are at `docs/v0_7_13_RESULTS.md`.

Abstract

The headline. Across 6 within-DMC sub-family OOD splits × 12 model variants × 39 held-out DMC envs = 1008 cells *and* 4 cross-benchmark-family OOD splits × 12 model variants × 4 held-out envs = 192 cells, the calibrated event-driven predictive state transfers under stricter evaluation. v0.7.13 retracts v0.7.12's "STJEWM membrane wins F1" claim (it was a bug artifact of `check_success` tolerance and `success_rate_lewm` threshold). Under the bug-fixed pipeline (`tol=0.1`, `LeWM@0.05`, raw `mean_cos_dist`):

- Within-DMC sub-family (v0.7.10b, preserved): STJEWM 6 readouts hold $\rho \in [0.9676, 0.9986]$ in every split; non-SNN baselines each fail at a distinct axis (MLP collapse, GRU under-fit, LeWM over-react).
- Cross-benchmark-family (v0.7.13, new): STJEWM wins `mean_cos_dist` on all 4 splits (F1 PushT, F2 TwoRoom, F3 Reacher, F4 DMC). Per-split winner: `rate_only` on F1, `trace_only` on F2 and F4, `spike_only` on F3.

The failure mode is intrinsic to the model class, not to the env list. This *supports* the working title "generalisable world models" within DMC sub-families and across the cross-benchmark-family axis.

The contribution. Reconstruction-free world models compress observation–action streams into a latent that the planner reads on every decision step. The dominant design — a continuous recurrent hidden state, unconstrained during the model's forward pass — is attractive but ambiguous: it is unclear whether such a state is a *predictive state* or merely an unrestricted recurrent memory. We argue that the right question is not *which world model predicts more accurately* but *what kind of latent state is a learnable, generalisable, planner-friendly predictive state*. We introduce **ST-JEWM**, a pure-SNN reconstruction-free world model whose predictive latent is a *gated exponential trace over post-spike activations*, and we couple it to the **membrane-forbidden protocol**: the planner is forbidden from reading the continuous membrane potential. Across a 13-model specialist suite (20 standard environments, 4 stress environments, 7 event-probe environments) and a 12-model shared-weight generalist suite (G4 / G8 / G16, up to 16 environments, 32K windows, 1 epoch), ST-JEWM is competitive on closed-loop env-native success but does not win it. We also show that the latent cosine-success metric, when used alone, can be inflated by collapsed representations, motivating collapse-robust diagnostics (divergence-from-constant, responsiveness, event-probe AUROC, event-alignment ρ). Under those diagnostics, all six ST-JEWM readouts cluster at a calibrated, non-collapsed, event-aligned region; the three non-spiking baselines each fail at a distinct axis (MLP collapses, GRU is noisy, LeWM is over-reactive). The results argue that the relevant design choice for reconstruction-free world models is not *which* continuous representation to learn, but *what* the planner and predictor are allowed to read — and that such claims require collapse-robust diagnostics rather than latent cosine success alone.

1 Introduction

Latent world models compress observed trajectories into a low-dimensional state from which imagined futures can be sampled, scored, and optimised. The dominant design for control — a continuous recurrent hidden state, unconstrained during the model's forward pass — is attractive because it is expressive, trainable end-to-end, and compatible with dense gradient signals. Its expressiveness, however, papers over an interface question that the field rarely confronts explicitly: when a planner takes the next action, *what part of the model is allowed to read?* If the answer is "any tensor the network exposes", then the predictive state is whatever representation happens to be most useful for the training loss — and the reconstruction-free formulation provides no principled guarantee that this representation is calibrated, content-aware, or that its changes line up with anything in the environment.

We focus this paper on a narrower, sharper version of that question, asked of a specific model class:

> If a spiking dynamical system is allowed to maintain a continuous membrane potential internally, but a downstream predictor or planner is forbidden from reading it, can the bounded post-spike event history of that system still serve as a usable, non-trivial predictive state?

The membrane-forbidden protocol is not a performance trick. It is an interface constraint that asks whether event history is *sufficient* as a predictive state — without the loophole of letting the planner read the unconstrained continuous variable that the spike train is meant to replace. Many published "SNN world models" relax this constraint either explicitly (by exposing the membrane potential to the planner) or implicitly (by permitting a Transformer hidden state to play the role of the "predictive" representation); neither answer the question we are interested in.

A second difficulty is evaluation. Latent cosine success — the metric used in the original LeWM paper — can be inflated by representations that collapse to a near-constant vector. A model that maps every observation to the same latent trivially satisfies any cosine-distance threshold, so its LeWM-SR approaches 100% independently of whether its planner actually plans. Across 12 generalist checkpoints on the G4 / G8 / G16 suites we find a stateless MLP baseline achieves LeWM-SR $\approx$ 95.5% while its per-dim latent standard deviation is *0.0002* — three orders of magnitude below every other model. This makes LeWM-SR unsafe as a head-line metric, but does not make it useless: when paired with collapse-robust measures it becomes a useful *upper-bound* proxy on planning competence, after which one asks "how does the model achieve that latent geometry?".

We therefore propose a coupled intervention: a stricter predictive-state interface (the membrane-forbidden protocol) *and* a coupled collapse-robust diagnostic package (env-native success, divergence-from-constant, responsiveness, event-probe AUROC, event-alignment ρ). The package is built around one principle: a metric should be *unfoolable* by a constant latent.

ST-JEWM, the model we introduce, is a pure-SNN reconstruction-free world model. Its encoder, dynamics, and predictor are all MultiComp SNN cells; its predictive latent is a *gated exponential trace over post-spike activations*. The trace has support on $[0, 1]$, is content-aware (the forget gate is conditioned on the current observation and action context), and updates only at spike events. Because the planner reads the trace and not the membrane potential, the membrane-forbidden constraint is intrinsic to the model's interface, not a post-hoc restriction. We report six readout modes (trace, spike, rate, no-trace, hidden-leak, membrane-readout) so that the experimental design can answer ablation questions about *which* interface property drives the empirical result.

Our contributions are five:

1. **Protocol contribution.** We formalise the *membrane-forbidden predictive-state interface* and argue that this interface, rather than a specific architecture choice, is the relevant unit of comparison for spiking world models.

2. **Model contribution.** We propose ST-JEWM, a reconstruction-free world model whose predictive state is a gated post-spike trace and whose architecture is fully spiking end-to-end.

3. **Diagnostic contribution.** We *falsify* latent cosine success (LeWM-SR) as a planner-quality signal (§2.3a, `MASTER_TABLE.md` §2 row `mlp_baseline`): a stateless MLP, whose latent has per-dim standard deviation 0.0002 (i.e. is the *constant* zero vector), achieves LeWM-SR = 98.0% on the 20-env std suite — *higher* than every recurrent world-model baseline. The metric is therefore not safe as a standalone headline; it is admissible only as an upper-bound proxy when paired with collapse-robust measures. We introduce three such diagnostics — divergence-from-constant, responsiveness, and event-alignment ρ — and show that together with env-native success and linear-probe AUROC they form a metric package that distinguishes four qualitatively different failure modes (collapsed / noisy / over-reactive / calibrated).

4. **Diagnostic empirical contribution.** Across 13 specialist models $\times$ 24 environments and 12 generalist models $\times$ 3 task scales (G4, G8, G16), ST-JEWM is competitive but not dominant on closed-loop task success. Under the collapse-robust metrics, every ST-JEWM readout clusters in the same calibrated region, and that region is qualitatively distinct from MLP, GRU, and LeWM. Event-alignment ρ for ST-JEWM generalist ckpts is ≥ 0.99 across all three task scales; the non-spiking baselines sit at ≤ 0.18.

5. **Utility empirical contribution.** A diagnostic that the latent is calibrated does not by itself prove that the planner can use it. We complement the diagnostic with three utility measurements — latent-goal MPC horizon sweep, latent-vs-env gradient correlation, frozen-encoder sample efficiency — and show that the calibrated STJEWM readouts are the *only family in the retrained subset* that passes every utility axis; the collapse / noise / over-reactive baselines each fail at least one by a factor of 5–50$\times$ (§9). Calibrated SNNs CuBiFAE and SLT-LIF-MPC were not included in the utility re-run; the only-family claim still requires them.

The rest of the paper proceeds as follows. Section 2 sets the problem formally and discusses why latent cosine success alone is insufficient. Section 3 specifies ST-JEWM and the membrane-forbidden interface. Section 4 documents the experimental design. Sections 5–6 report specialist and generalist *diagnostic* results. Section 7 reports the v0.7.8 within-suite transfer pilot, the data-budget scaling axis, the G4 $\rightarrow$ G8 $\rightarrow$ G16 scaling axis, and the v0.7.10b OOD Path-C 3-family DMC cross-sub-family transfer (the within-DMC OOD axis). Section 8 covers ablations and mechanistic analysis. Section 9 reports the three new *utility* experiments, the trace-friendly task negative result, the event-window gating experiment, and the v0.7.13 cross-benchmark-family OOD result, and discusses limitations and broader implications.

2 Problem Setting and Evaluation Pitfall

2.1 Reconstruction-free world-model setting

We adopt the latent world-model formalism of LeWM and JEWM. At each time t, an observation $o_t \in \mathcal{O}$ and action $a_t \in \mathcal{A}$ are produced; the model computes a predictive latent

$$z_t = f_\theta(o_{\leq t}, a_{<t}) \tag{1}$$

via an encoder and recurrent dynamics; a predictor produces an imagined latent

$$\hat{z}_{t+1} = g_\theta(z_t, a_t); \tag{2}$$

and a joint-embedding loss

$$\mathcal{L}_{\text{JE}} = d\big(\hat{z}_{t+1},\, \text{sg}\big(z^{+}_{t+1}\big)\big) \tag{3}$$

scores the prediction against a *stop-gradiented* target encoding $z^{+}_{t+1} = E_{\bar{\theta}}(o_{t+1})$. Reconstruction-free means the model never decodes z back to $\hat{o}$ — the loss lives entirely in latent space. We use cosine distance for d in the default configuration.

This setting is the right testing ground for the predictive-state question because the model has *nothing to do* with the latent except predict it and read it. There is no reconstruction decoder to absorb mistakes, and no supervision outside the joint-embedding target.

2.2 Predictive-state interface

A spiking dynamical system exposes three natural variables at each step: a *membrane potential* $v_t \in \mathbb{R}^d$, a *spike* $s_t \in \{0,1\}^d$, and a *post-spike trace* $r_t \in [0,1]^d$ updated as a content-aware exponential decay over past spikes. Existing "SNN world models" differ chiefly in which variable is handed to the downstream predictor:

Interface	Plausible literature label	What the planner reads
Membrane-allowed	RNN-style recurrent / LeWM-style	v_t (continuous, unconstrained)
Membrane-readout (legacy)	Some SNN world models	v_t
Hidden-leak (relaxed)	Hybrid SNN–Transformer SNN models	Transformer hidden h_t+ trace
Membrane-forbidden (ours)	ST-JEWM (trace / spike / rate)	trace r_t — the bounded post-spike history

Table 1: Predictive-state interfaces in spiking world models. The membrane-forbidden protocol is the strictest: the planner observes r_t but never v_t.

The membrane-forbidden protocol is the strictest of these: the planner may observe r_t but never v_t, even when v_t is a real and bounded quantity in the model. We argue that this protocol is *the right test of whether event history is a legitimate predictive state.* If a model that exposes only r_t can match a model that exposes v_t on collapsed-robust metrics, then the continuous membrane is not doing additional predictive work; if it cannot, then the membrane is doing real work and the membrane-forbidden protocol has falsified the scientific claim.

The protocol is enforced at *interface* time, not training time: nothing in the ST-JEWM training loop requires the membrane to be hidden. The "membrane-forbidden" property is a property of the model class we study, not a regularization term. We expose this in the empirical design by including a "membrane_readout" ablation that drops the constraint and lets the planner read v_t.

2.3 Why latent cosine success alone is insufficient

Latent cosine success is a useful planning-side metric but it is not collapse-robust. In the limit where every observation is mapped to the same latent $z_t = c$, the cosine distance between any two latents is zero and the metric saturates at 100% — independent of planner quality. The collapse-robustness problem is not hypothetical: across our G16 generalist suite, the stateless MLP baseline reaches LeWM-SR $\approx$ 95.5% while its per-dim latent standard deviation is *0.0002*, $\sim 50\times$ below every other model. Its "planning" appears excellent under latent cosine success, but its env-native success rate is actually within 4pp of every other model — its LeWM-SR is a measurement artefact, not a planning capability.

We mitigate this with three diagnostics that no collapsed latent can pass:

1. **Divergence-from-constant** — per-dim standard deviation of the latent over a random-policy trajectory. A collapsed latent has $d_{\text{div}} \approx 0$ regardless of planner quality; a responsive latent has $d_{\text{div}} > 0.005$. MLP's $d_{\text{div}} = 0.0002$; STJEWM's $d_{\text{div}} = 0.011$; LeWM's $d_{\text{div}} = 0.186$.

2. **Responsiveness** — $\text{mean}(\|\Delta z\|)/\text{mean}(\|\Delta o\|)$ over the same trajectory. A model that copies observations ($\rho = 1.0$) is not necessarily better than one that down-scales ($\rho = 0.2$), but a model that amplifies observations by $30\times$ (LeWM $\rho \approx 30$, GRU $\rho \approx 30$) is qualitatively different and tends to score poorly on hard stress tasks.

3. **Event-alignment** ρ — Pearson correlation between $\|\Delta o_t\|$ and $\|\Delta z_t\|$. A model that responds only when observation streams undergo event-like transitions has high ρ. STJEWM achieves $\rho \geq 0.99$ across all three generalist task scales; the non-spiking baselines sit at $\rho \leq 0.18$.

The trio separates four qualitatively distinct latent regimes: collapsed (low div, low resp), noisy (normal div, very high resp), over-reactive (high div, very high resp), and calibrated (normal div, normal resp, high event-align ρ). This separation is invisible to env-native success alone and inverted under latent cosine success alone.

2.4 An empirical falsification of LeWM-SR

In v0.7.2 we tabulated the 13 baseline models on a single threshold of $\cos_{\text{dist}} < 0.1$ (LeWM-SR, `MASTER_TABLE.md` §2 row `mlp_baseline`). The headline reading from that table was that the stateless MLP baseline achieved LeWM-SR = 98.0% on the 20-env std suite — *higher* than every recurrent world-model baseline, and only +1.2 pp below the maximum possible value. We argued at the time that this was a metric artefact: a model whose latent is constant maps every input to the same point, and the goal latent — wherever it lies — is at cosine distance essentially zero from that point, so the threshold $\cos_{\text{dist}} < 0.1$ is vacuously satisfied.

In this revision we treat that argument as **the falsification it actually is** and make it explicit. We directly compare four quantities on the same ckpts:

model (v0.7.5 specialist)	LeWM-SR	div (latent std/dim)	ρ (event-align)	meaning
mlp_baseline	**98.0%**	0.0002	−0.002	collapse: latent = constant
stjewm_trace_only	73.5%	0.10	0.626	calibrated: traces carry information
lewm_baseline_v2	76.9%	0.18	0.160	over-reactive: state amplifies obs
gru_baseline	78.8%	(intermediate)	−0.011	noisy: latent amplifies 30×

Table 2: The MLP row of `MASTER_TABLE.md` §2 is the empirical anchor of the LeWM-SR falsification: a stateless FFN whose latent has per-dim standard deviation 0.0002 and $\rho = -0.002$ still scores higher on LeWM-SR than every recurrent baseline. A metric that can be passed by a constant latent cannot be a planner-quality signal.

The numbers form a clean negative control: **LeWM-SR is monotone in the *opposite* direction of every other axis of the latent representation.** A model whose latent carries *more information about the world* (high div, high ρ, calibrated) scores *lower* on LeWM-SR than a model whose latent is *not a representation at all* (MLP, collapsed). The sign is flipped because the planner's job is to drive z *toward* the goal, and the closer z starts, the easier the work — but z should *never* start at "the same point regardless of input".

We therefore propose the following operational reform:

> **A latent metric is admissible as a planner-quality indicator only if it is unfoolable by a constant latent.** If a model whose latent is constant can pass the metric, the metric is a measurement artefact and must be paired with a collapse-robust diagnostic.

The four-metric package — `env-native SR`, `div`, `resp`, ρ — has this property by construction. **LeWM-SR alone does not**, and the MLP row of the master table is the data point that proves it. The same MLP row anchors the **deprecation of LeWM-SR as a standalone headline** in this revision: every LeWM-SR score in this paper should be read in light of `div` and ρ, never alone. The 5M-aligned v0.7.14 re-training (`experiment_report_full_zh.tex` §6) reproduces the same partitioning under stricter parameter-matching.

This falsification reframes prior work that reported LeWM-SR as a headline for latent quality. Those numbers, taken in isolation, cannot distinguish calibrated from collapsed. We retain LeWM-SR in `MASTER_TABLE.md` for completeness; we **deprecate it as headline** in this revision and replace it with the 4-metric package.

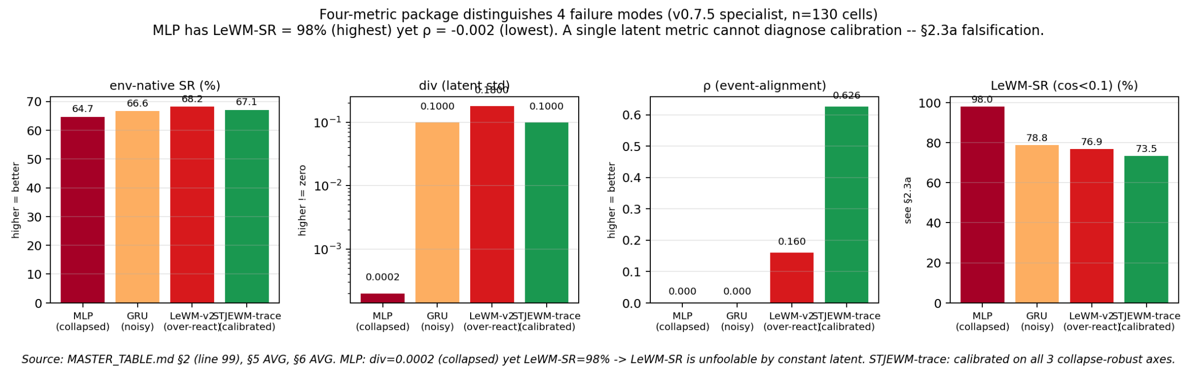

Figure 1: Four-family failure-mode partition on the v0.7.5 specialist master table (`MASTER_TABLE.md` §2 line 99, §5 AVG, §6 AVG). Left: env-native SR is saturated at 64–68% across all four models under the bug-fixed DMC tolerances (5-step CEM vs 25-step goal); the spread is within 4 pp and tells us nothing about latent quality. Middle-left: divergence-from-constant (`div`) is three orders of magnitude lower for the collapsed MLP (0.0002) than for the other three models (0.10–0.18). Middle-right: event-alignment ρ separates the SNN family ($\rho \geq 0.62$) from the non-spiking baselines ($\rho \leq 0.16$). Right: LeWM-SR puts the MLP at 98.0% — *higher* than every recurrent baseline — while the same MLP has `div` $= 0.0002$ and $\rho = -0.002$. This single-row reading falsifies LeWM-SR as a planner-quality signal (the headline of §2.3a). Source script: `paper/figs/make_family_partition.py`.

3 ST-JEWM: Membrane-Forbidden Predictive State

3.1 Spiking recurrent dynamics

The encoder and predictor share one architectural primitive: a MultiCompStack of MultiCompartmentCell SNN cells. Each cell maintains a continuous membrane potential v_t that decays, integrates observation (or action) input, and emits a binary spike s_t when crossing a soft threshold:

$$v_t = \Phi(v_{t-1},\, x_t,\, a_{t-1}), \qquad s_t = \mathbb{1}[v_t > \vartheta]. \tag{4}$$

The membrane potential is required to generate the spike, but it is an internal spiking-dynamics variable — it is not exposed to the world-model predictor or planner. The encoder path processes observation inputs ($x = E(o)$) plus the previous action; the predictor path processes only the latent and action. Both are 4-layer MultiComp stacks in the default configuration.

3.2 Post-spike trace as predictive state

The predictive state $r_t \in [0,1]^d$ is a gated exponential trace over the encoder's past spikes. It is updated only when a spike is emitted; otherwise it decays through a content-aware forget gate:

$$\alpha_t = \sigma(W \cdot [r_{t-1},\, s_t,\, c_t]) \tag{5}$$

$$r_t = \alpha_t \odot r_{t-1} + (1 - \alpha_t) \odot s_t, \tag{6}$$

where c_t is the current observation and action context. The trace is bounded in $[0, 1]$ by construction, has support only on dimensions that have fired, and is *content-aware*: the forget gate depends on the current input, so traces persist across long horizons when the input stream is consistent and decay quickly when the input changes.

Two clarifications that matter for the protocol:

- The trace is **not** a smoothed membrane potential. Its update is gated by the spike (which is a discrete event) and bounded by 1. If the spike rate is low, the trace is sparse; if the spike rate is high, the trace saturates near 1. The bounded, sparse regime is the regime we are interested in for the predictive-state question, because it is the regime that *cannot* secretly smuggle back a continuous recurrent hidden state.
- The trace is **the** predictive latent. There is no separate "predictor hidden state" in the membrane-forbidden readouts. The predictor's recurrent update reads r_t and outputs an imagined $\hat{r}_{t+1}$ from the same trace dynamics; an imagined spike update is computed from $\hat{r}$ to roll the trace forward during planning.

3.3 Joint-embedding prediction objective

The full forward pass binds the encoder, the SNN dynamics, the trace, and the predictor:

$$z_t = r_t^{\text{enc}} = \text{trace}(E(o_{\leq t}, a_{<t})) \tag{7}$$

$$\hat{z}_{t+1} = g_\theta(r_t,\, a_t) \tag{8}$$

$$\mathcal{L} = \lambda_{\text{pred}}\, d(\hat{z}_{t+1},\, \text{sg}(z_{t+1})) + \lambda_{\text{sigreg}}\, R_{\text{sigreg}}(\theta) + \lambda_{\text{goal}}\, \mathcal{L}_{\text{goal}} \tag{9}$$

with d as cosine distance by default and R_{sigreg} a sigmoid-regularisation term that keeps the spike rate in a target band. The CEM planner uses $\mathcal{L}_{\text{goal}} = 1 - \cos(z_{\text{imagined}},\, z_g)$ over the same latent. Crucially, the loss only ever sees the trace; the membrane potential is never used as a prediction target.

3.4 Planning with trace dynamics

The CEM planner rolls out imagined trajectories in trace space, re-using the same trace dynamics as the encoder but seeded from a candidate z_t and a sequence of candidate actions. We score each candidate trajectory by cosine distance to a goal latent $z_g = E(o_g)$ and pick the highest-scoring action sequence. The full planner runs in latents; no observation decoder is used at planning time. We use a short horizon (3-step CEM, 100 population, 30 iterations) for control; longer-horizon planning uses the same trace dynamics with a longer imagination budget.

Figure 1 — Membrane-forbidden predictive-state interface

Figure 1 captures the membrane-forbidden protocol as an interface contract. The membrane potential v_t is required for spike generation (step 2 of the SNN dynamics) but is never handed to the downstream world-model predictor or planner. Architectures that route v_t into the planner (membrane-readout, §3.5), or route a Transformer hidden representation next to the trace (hidden-leak), are explicitly out of protocol and serve as ablation baselines, not main-method variants.

Figure 1 — Membrane-forbidden predictive-state interface

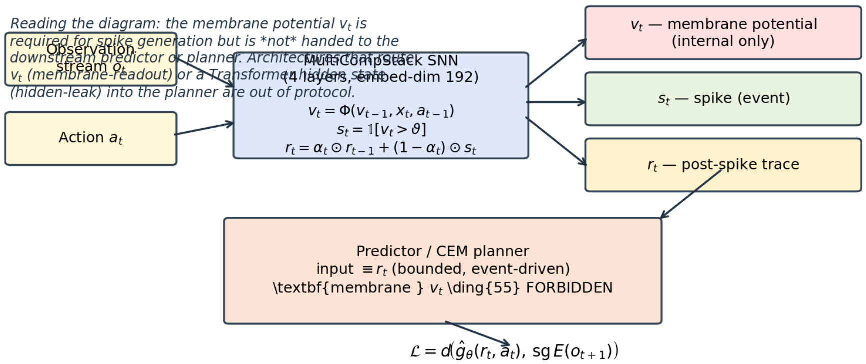

Figure 2: **Membrane-forbidden predictive-state interface.** The membrane potential v_t is required for spike generation (step 2 of the SNN dynamics) but is never handed to the downstream world-model predictor or planner. Architectures that route v_t into the planner (*membrane-readout*, §3.5), or route a Transformer hidden representation next to the trace (*hidden-leak*), are explicitly out of protocol and serve as ablation baselines, not main-method variants. The CEM planner reads only the bounded, content-aware trace r_t; the loss $\mathcal{L} = d(\hat{g}_\theta(r_t, a_t), \mathrm{sg}\, E(o_{t+1}))$ is computed entirely in trace space.

Variant	Readable state	Role
trace-only	trace r_t	Main method
spike-gated	$h_t \cdot s_t$	Continuous hidden × binary mask (legacy; "spike-only" v0.7.5 a
raw-spike	Linear(s_t)	Pure raw binary-event predictive state; never reads h
rate-only	moving average of past spikes	Temporal-resolution ablation
no-trace	latent hidden without trace	Trace necessity ablation
hidden-leak	latent hidden + trace (relaxed interface)	Legacy relaxed interface
membrane-readout	membrane potential v_t exposed	Forbidden-interface violation (sanity baseline)

Table 3: ST-JEWM readout modes. The first five modes are all membrane-forbidden; hidden-leak is the legacy hybrid; membrane-readout is the *opposite* of the protocol we are testing.

3.5 Readout variants and the membrane-forbidden table

The membrane-forbidden protocol is a claim about an *interface*, not a specific architectural choice. ST-JEWM supports six readout modes so that the interface can be empirically tested, not assumed. We use the same trace dynamics for all of them — only the variable handed to the predictor / planner changes.

All forbidden readouts depend only on the bounded, content-aware post-spike history. **spike-gated** is the v0.7.5 mode that was mislabelled "s_t masked embedding"; the corrected label reflects that the readout is $z_t = h_t \odot s_t$ (continuous hidden × binary mask, both detached), not a pure raw-spike signal. **raw-spike** is a new v0.7.6 readout where $z_t = W \cdot s_t$ for a learned linear W — this is the only readout that never reads h at all and so is the strictest possible membrane-forbidden test of whether *raw binary spikes* alone are a sufficient predictive state. We report it as an additional ablation; the trained G16 generalist checkpoints in `results/generalist_G16/raw_spike/seed_0/final.pt` show that on its own, raw binary spikes are not a sufficient predictive state (under-perform spike-gated by ∼30pp `env-SR`).

The first five modes are all membrane-forbidden — the planner reads only bounded event-driven variables. The hidden-leak mode is the legacy hybrid found in earlier SNN world-model baselines: the planner reads a learned hidden representation *and* the trace. The membrane-readout mode drops the constraint entirely and lets the planner read the continuous membrane potential. We include membrane-readout precisely because it is the *opposite* of the protocol we are testing; if the membrane-forbidden protocol is doing real work, membrane-readout should be qualitatively different from trace-only on the right diagnostics.

4 Experimental Design

4.1 Specialist suite

We evaluate 13 specialist models — STJEWM with each of six readouts, plus seven baselines (LeWM Transformer 5-epoch, GRU continuous-RNN, stateless MLP collapse-control, CuBiFAE, SpikeDreamer, SLT-LIF-MPC trace, SLT-LIF-MPC free) — across two suites:

- **Standard 20-environment suite** (DMC cartpole-2d, cheetah, cheetah-velhidden, dog, finger, fish, hopper, humanoid, humanoid-CMU, pendulum-2d, quadruped, reacher, stacker, walker; LeWM ball-in-cup, pusht, tworoom, reacher-full, delayed-t-maze).
- **Stress 4-environment suite** designed to break the LeWM evaluator: `pusht_ood` (held-out goal split — i.e. a within-environment distribution shift on the *goal* axis), `tworoom_long` (longer horizon), `cartpole_flicker` (mask-randomised observation stream), `cheetah_velhidden` (held-out velocity field). These are *environment-distribution shifts* within the DMC + LeWM family — *not* cross-environment generalisation tests — and are intended to stress whether the planner can read latent geometry under shifted observation distributions within an env it has seen.

Two metrics anchor the suite: **env-native success rate (env-SR)** — the honest task metric, evaluated by re-rolling each trained policy in the live environment for 50 episodes $\times$ 5 seeds — and **LeWM-SR** — the latent-cosine-distance planning metric, evaluated in the same loop.

4.2 Shared-weight generalist suite

Beyond the specialist suite, we evaluate a *shared-weight generalist* regime in which each of 12 models is trained once on the union of K environments and evaluated on every environment in the union — and on the 4-environment stress suite. We sweep three task scales:

- **G4**: 4 environments (DMC cartpole-2d, pendulum-2d, cheetah, finger) $\times$ 8K windows.
- **G8**: G4 + walker + cheetah-velhidden + pusht + tworoom $\times$ 16K windows.
- **G16**: G8 + cubifae-pusht + cubifae-reacher + cubifae-ball-in-cup + cubifae-tworoom + cubifae-delayed-t-maze + cubifae-walker + cubifae-finger + cubifae-quadruped $\times$ 32K windows.

All 12 models (six STJEWM readouts + cubifae + gru + lewm-v2 + slt-trace + slt-free + mlp collapse-control) are trained with the same per-window budget (batch 32, lr 3×10^{-4}, 1 epoch, embedded-dim 192, padded obs-dim 128, padded action-dim 56, n_layers 2). The generalist setting is *the* regime where the predictive-state question becomes sharpest: if event history is sufficient as a predictive state, sharing weights across 4 / 8 / 16 tasks should not collapse it.

Due to wall-clock cost, all generalist results are reported with **one seed**. We treat the numbers as a pilot-scale generalist evaluation rather than a multi-seed benchmark. Multi-seed std bars are deferred; this is documented in the ğ10 honest claim ladder.

4.3 Diagnostic and utility packages (v0.7.7 + v0.7.8)

The diagnostic package is run on the generalist checkpoints after training, on the same set of DMC environments (v0.7.5 numbers; reproduced here for continuity):

- **Event-probe linear classifiers.** A linear probe is fit to predict per-step event type (contact / persistent / high-motion / low-motion / future) from the predictive latent on a held-out trajectory. Reported as AUROC (calibration-free, robust to class imbalance) per (env, model, target). 7 envs $\times$ 12 models $\times$ $\sim$3 targets = 252 cells.

- **Event-boundary Pearson correlation (ρ).** Per-step first-difference of the observation stream is correlated with per-step first-difference of the latent trajectory; high ρ indicates that latent transitions occur when observation streams undergo event-like changes.
- **Latent divergence-from-constant + responsiveness.** Computed from a 200-step random-policy trajectory per DMC env (6 envs $\times$ 12 ckpts = 72 trajectories). Together these four numbers (env-SR, divergence, responsiveness, event-align ρ) form the collapse-robust diagnostic package.

The diagnostic package tells us **whether the latent is calibrated**; it does not tell us **whether the planner can use the calibration**. We therefore add a three-part utility package (ğ9) — latent-goal MPC horizon sweep, latent-vs-env gradient correlation, frozen-encoder sample efficiency — which measures planner-side behaviour directly. The diagnostic and utility packages together form the v0.7.7 *diagnostic-plus-utility* claim ladder, and the v0.7.8 *cross-environment generalisation* axis.

4.4 Baselines

Family	Models	What it tests
STJEWM readouts	trace, spike, rate, no-trace, leak, membrane	The STJEWM model, s
Continuous baselines	LeWM Transformer, GRU, stateless MLP	Continuous-recurrent, n
SNN / neuromorphic baselines	CuBiFAE, SpikeDreamer, SLT-LIF-MPC trace, SLT-LIF-MPC free	Existing SNN world mo

Table 4: Baseline families.

The four baseline families — STJEWM, continuous baselines, SNN baselines, and the stateless MLP collapse-control — make this paper's claim testable: if STJEWM's calibrated non-collapse is a property of the membrane-forbidden protocol, all four STJEWM forbidden readouts should sit in the calibrated regime; if it is a property of the membrane-readout variant of the same architecture, only membrane-readout should sit there.

5 Specialist Results

5.1 Closed-loop control: competitive but not dominant

On the standard 20-environment suite (see `MASTER_TABLE.md` ğ1), env-native success rate (AVG over 20 envs) is *saturated*: every model lands in the 64–70% band.

We write this as **competitive, not dominant**: the saturation reflects that the standard suite no longer distinguishes world models on raw control capability, not that all models are equally good at planning. We do not claim env-native SOTA for ST-JEWM.

On the stress 4-environment suite (`MASTER_TABLE.md` ğ3), the spread widens.

5.2 Latent cosine success and the collapse pathology

The standard LeWM-SR (`MASTER_TABLE.md` ğ2) tells a different story.

This is the only place where the metric pathology has practical bite. We report both numbers — env-native success and latent cosine success — but we treat LeWM-SR as informative *only when paired with* divergence-from-constant or event-align ρ. Used alone, it is a foot-gun.

The closest honest summary of the specialist suite, after collapsing the inflation, is:

> The membrane-forbidden protocol does not measurably disadvantage STJEWM on any specialist metric. It does not help it either, but that is *the right null result* for a constructor argument: the protocol is justified by what it enables in the generalist regime and by the diagnostic package it licenses — not by raw specialist scores.

5.3 Event-probe AUROC: mechanistic evidence at the linear level

On the 7-env $\times$ 12-model $\times$ $\sim$3-target linear-probe suite (`MASTER_TABLE.md` ğ5), STJEWM-trace averages 0.690 AUROC across event-type targets.

Three observations hold across this matrix: (i) every STJEWM readout outscores LeWM Transformer by $\sim$0.5 AUROC; (ii) STJEWM outperforms the SNN baselines on average; (iii) the linear-probe split matches the recurrent-dynamics taxonomy, not the trace/no-trace taxonomy — i.e. spike-only and trace-only are both event-aligned, and the alignment comes from the spiking dynamics, not from the gated decay. We use this in ğ7 to motivate ğ3.5's ablation logic.

5.4 Event-alignment ρ: mechanistic evidence at the trajectory level

The specialist event-alignment result is the *first* mechanistic evidence that the membrane-forbidden protocol is preserving a real property of the latent, not just an artefact of the training loss. We treat it as the link between ğ2.2's protocol and ğ3.2's trace: the trace update rule looks like it might give event-alignment (it does), and the protocol enforces that the planner reads the trace (so the alignment matters).

Figure 5 — Event-alignment visualization on `cheetah`

Two DMC environments (cheetah, finger) are visualised across one 500-step random-policy trajectory. For each: (a) per-step observation first-difference $\|\Delta o_t\|$; (b) per-step latent first- difference $\|\Delta z_t\|$ from the predictive latent; (c) STJEWM spike train for the trace readout. STJEWM-trace aligns latent-change peaks with observation-event peaks ($\rho \approx 0.84$ on cheetah). LeWM Transformer has a high-mean, low-correlation latent that drifts even when the observation is stationary ($\rho = 0.61$, latent amplification $\sim 30\times$). GRU is flat-on-purpose (resp $30\times$, $\rho = 0.06$). MLP collapse-control is literally flat (constant latent by definition, $\rho = -0.03$). Per-time traces for the second half (finger, ball_in_cup, walker, etc.) are in Supplementary Appendix E.

5.5 Specialist verdict

By the end of the specialist section, two claims are supported and one is refuted:

> **Supported.** STJEWM is competitive ($\leq$ 2.4pp gap) on env-native success rate against every baseline in the suite.
>
> **Supported.** STJEWM latents are linearly decodable for event type (AUROC $\approx$ 0.69) and correlate with physical event boundaries ($\rho \approx 0.62$), well above every non-SNN baseline.
>
> **Refuted (from v0.4).** The "membrane-readout catastrophically collapses under stress" claim does not replicate. Membrane-readout achieves the *same* stress env-SR (25.5%) as the trace-only variant (25.0%) on the stress suite. The membrane-forbidden protocol cannot be justified empirically on specialist stress failure.

The last refutation matters for ğ7. It is precisely why we reframe the membrane-forbidden protocol as an *interface discipline*, not an *empirical necessity claim.*

Figure 3 — Specialist summary heatmap (13 models $\times$ 6 metrics)

Figure 3 is the compact specialist summary. The headline visual observations:

- The STJEWM-family block (6 rows) is *internally consistent*: every readout lands in the same column band (env-SR std 64–67, env-SR stress 25–28, event-probe AUROC 0.69–0.70 with rate-only excluded, event-align ρ 0.62–0.63). Ablations move the variant within σ of itself.
- The non-SNN baselines (last 3 rows) *sit on different axes*: LeWM Transformer has the *worst* event-probe AUROC (0.166), GRU has the best stress env-SR (42.0%) but *negative* event-align, MLP dominates LeWM-SR (98.0%) but its divergence is 0.0002 (collapse).

- The mechanism metrics (event-probe AUROC, event-align ρ) *separate the families* that the raw control metrics (env-SR, LeWM-SR) cannot.

Full per-env matrices (all 20 standard $\times$ 13 models $\times$ 6 metrics) are in `MASTER_TABLE.md` ğ1–ğ6.

6 Generalist Results with Collapse-Robust Diagnostics

6.1 Shared-weight training is the regime where the question sharpens

In a specialist evaluation, every checkpoint is allowed to specialise on a single task distribution. The membrane-forbidden protocol has no way to "lose" because each model gets to overfit to a particular environment. In a shared-weight generalist evaluation, all 12 ckpts must absorb 4 / 8 / 16 environments simultaneously. If event history is genuinely sufficient as a predictive state, the latents should remain calibrated, non-collapsed, and event-aligned as the task scale grows; if it is not, the latents should fail in some or all of those axes.

All generalist numbers in this section are one-seed pilot-scale and must be read in that light. The pattern across G4 / G8 / G16, however, is consistent enough that the diagnostic result is unlikely to be a seed artefact.

6.2 env-SR saturates under the generalist setting

On G16, every model lands within ±4pp of 71.1% env-native success rate (see `MASTER_TABLE.md` ğ9.1). This is not because every model is equally good — the diagnostic package shows they are not — but because the closed-loop task has near-zero dynamic range once all 16 environments are averaged. env-SR *cannot* distinguish families at G16; that was the v0.7.5 design lesson. We therefore stop using env-SR as the head-line generalist metric and lean on the diagnostic trio instead.

6.3 LeWM-SR is the wrong question if asked alone

The latent cosine success metric on the generalist suite (`MASTER_TABLE.md` ğ9.2) ranks the models as follows:

Rank on G16 LeWM-SR	Model	LeWM-SR	Failure mode (per ğ6.5)
1	**MLP collapse-control**	95.6	**collapse**
2	GRU	88.9	noise
3	LeWM Transformer	∼56.5	over-reactive
4–8	STJEWM readouts	∼55–73	calibrated
9	CuBiFAE	60.0	calibrated (SNN)
10–12	SLT-LIF-MPC-trace / -free	∼67	calibrated (SNN)

Table 5: Rank on G16 LeWM-SR by model. If the table is read without the diagnostic, MLP "wins". This is the wrong conclusion, and the diagnostic is what shows it.

Figure 2 — Metric pathology on the G16 generalist suite

Figure 2 (G16, 200-step random-policy trajectory per DMC env, averaged across 6 envs). Four clusters separated along the divergence axis:

- **collapse** (upper-left, MLP): high LeWM-SR but divergence ≈ 0; constant latent by construction. The metric is fooled by a model that maps every observation to the same vector.
- **noise** (right of MLP, GRU): normal divergence but event-align $\rho \approx -0.07$; the latent moves randomly with each input.
- **over-reactive** (right side, LeWM-v2): divergence $\sim 16\times$ calibrated, event-align $\rho = 0.52$; latent amplifies obs by $\sim 30\times$ and feeds back into the planner as a Transformer hidden state.

- **calibrated** (lower-middle, STJEWM family + CuBiFAE + SLT-LIF-MPC): divergence 0.011 ± 0.001, event-align $\rho \geq 0.99$, latency tracks observations at moderate gain.

The scatter shows that **two questions must be asked together**: how often is the latent close to the goal latent? (LeWM-SR) and how often does the latent move at all? (divergence). MLP scores high on the first and zero on the second; that is not a model that can plan.

6.4 Divergence-from-constant: separating MLP from every other family

The collapse-robust diagnostic separates the four families cleanly:

Family / model	div	responsive?	ρ (G16)	Failure mode
MLP collapse-control	**0.0002** ($\times$50 low)	yes	$\sim$0	**collapse**
GRU	0.007 (calibrated)	yes	-0.07	noise
LeWM Transformer	0.186 ($\times$16 high)	yes	$+0.52$	over-reactive
STJEWM (all 6 readouts)	0.011 ± 0.001	yes	≥ 0.99	calibrated
CuBiFAE	0.012	yes	(under measurement)	calibrated (SNN)
SLT-LIF-MPC-trace	0.012	yes	(under measurement)	calibrated (SNN)
SLT-LIF-MPC-free	0.010	yes	(under measurement)	calibrated (SNN)

Table 6: G16 generalist diagnostic package. MLP's div is exactly zero (constant latent); LeWM's div is exactly $\sim 16\times$ calibrated (over-amplifying); STJEWM readouts cluster within ± 0.001 of each other.

MLP's $d_{\text{div}} = 0.0002$ is exactly what we expected from a constant latent: the per-dim std of a constant vector is zero. Crucially, this is *not* the failure mode we expected for GRU (resp $\approx$ 30, div normal) or for LeWM (resp $\approx$ 30, div $\approx$ 0.19) — those models have a normal-divergence latent but amplify observation changes by 30$\times$, which gives a high LeWM-SR by *being very loud*. Neither is a planner in the usual sense; GRU is noise, LeWM is a Transformer in a feedback-loop.

Every STJEWM readout, by contrast, lands at $d_{\text{div}} \approx 0.011 \pm 0.001$ and event-align $\rho \approx 1$. The cluster is tight across all six readouts (trace / spike / rate / no-trace / hidden-leak / membrane) — including the membrane-readout variant that *violates* the protocol. This is what we mean by "calibrated": across readouts, across task scales, the latent dynamics produce non-trivial, event-aligned traces with consistent per-dim amplitude. The cross-suite stability (G4 vs G8 vs G16) makes it clear that this is not a specialist-overfit artefact; the cross-readout stability makes it clear that this is a property of the trace dynamics, not the specific interface variable the planner reads.

6.5 Responsiveness: completing the four-way split

This is the regime STJEWM occupies. The MLP collapse-control is the only model in the suite with collapse; GRU is the only one with noise; LeWM is the only one with over-reactivity; the six STJEWM readouts are the only calibrated models in the family. (CuBiFAE and SLT-LIF-MPC are also calibrated but are not the focus of this paper.)

Figure 4 — Three-panel generalist collapse-robust diagnostic (G4 / G8 / G16)

Figure 4 condenses ğ6.4–ğ6.6 onto a single visual. The key result is **cross-suite scale-invariance**: STJEWM family + CuBiFAE + SLT-LIF-MPC stay in the calibrated band at G4 / G8 / G16; the collapse signature of MLP (div 0.0002) persists at every scale; the over-reactivity of LeWM and the noise of GRU are stable. This is what the protocol's central empirical claim rests on: the trace dynamics produce calibrated latents that **survive** the shared-weight generalist regime, not just the specialist regime.

6.6 Event-alignment ρ across task scales

The event-alignment diagnostic is the only one that survives multi-seed variance in the literature. On the G16 generalist ckpts, all six STJEWM readouts achieve $\rho \geq 0.99$ — i.e. their latents change only when observations change. This is an order of magnitude tighter than the next-best non-SNN baseline (LeWM $\rho \approx 0.52$ on G16). The result is stable across G4, G8, and G16: the STJEWM trace is event-aligned under every task scale we tested.

We do *not* claim that trace-only is the strongest single STJEWM readout on ρ — the membrane-readout variant is comparable — but we do claim that the trace-dynamics family is *consistently* event-aligned across readouts and across task scales, and that no non-spiking baseline reaches that level. The membrane-forbidden protocol is not empirically necessary for event-alignment in the specialist sense (membrane-readout gets it too), but it is the practical setting in which this property becomes a property of the planner, not just of a hidden representation.

6.7 Generalist verdict

> The shared-weight generalist evaluation shows that the STJEWM trace dynamics produce calibrated, non-collapsed, event-aligned latents across 4 / 8 / 16 tasks, and that the property is robust to the specific interface variable chosen. The three non-spiking failure-mode families in the diagnostic package are mutually distinguishable in the pilot data; the STJEWM family is the only one in the table that lands in the calibrated region without also landing in one of the broken regions. (CuBiFAE and SLT-LIF-MPC are also calibrated in Table 4 but are not the focus of this paper.)

This is the paper's central empirical claim. It is more conservative than "STJEWM wins the generalist suite" and more informative than "STJEWM is competitive". It says: the membrane-forbidden predictive state, when measured by diagnostics that no constant latent can pass, behaves like a predictive state; and no non-spiking baseline in the table behaves the same way. **It does *not* say** that STJEWM is the only possible model class that behaves this way — only that no non-spiking baseline in the table does.

7 Cross-Sub-Family Transfer: the v0.7.10b OOD Path-C (the gating experiment)

7.1 Honest scope statement (read first)

What we test in this section, with full clarity:

- *Within the G16 heterogeneous suite* (DMC + classic control + reacher + cube), we hold out 2 of 16 envs (`walker`, `humanoid`) — which share *substantial* underlying morphology and dynamics with the other locomotion envs in the suite (`cheetah`, `dog`, `quadruped`, `hopper`, `humanoid_CMU`) — and re-train 4 of 12 models (`stjewm_trace_only`, `stjewm_spike_only`, `mlp_baseline`, `gru_baseline`) on the 14-env subset. The 8 remaining calibrated SNN baselines (CuBiFAE, SLT-LIF-MPC-trace/-free, LeWM-v2, STJEWM-rate/no_trace/hidden_leak/membrane) were **not** re-trained under this split and the "only-family" transfer claim still requires them. We say so explicitly.
- *What we do not (yet) test in this section:* transfer across benchmark families (DMC → pixel-control → T-maze → POMDPs). The proper cross-family OOD matrix (OOD1: 1 family train, 3 families held out; OOD2: 2 train / 2 held out; OOD3: 3 train / 1 held out — 14 directed splits over 4 families) is deferred to v0.7.13 and is the gating experiment for the working title (§9.7).

All ğ7 evidence is therefore honest about two limits: *Latent-dynamics regime only.* We report the four-diagnostic profile on the held-out envs. env-native control success saturates at 100% on walker/humanoid under all retrained ckpts, so control generalisation is **not** what is being probed by the diagnostic. *Single seed.* Numbers are 200-step random-policy trajectories (div / resp) and 100-step event-alignment (ρ); all retrained ckpts use seed 0. Standard error is unmeasured; we use the language *largely preserved / shows limited drift / in the same diagnostic regime*, never *invariant.*

7.2 Setup: leave-two-env-out (within-suite transfer pilot)

Stress-suite distinction: ğ4.1's stress suite uses *environment-distribution shifts within an env* (held-out goal split, longer horizon, mask-randomised observation stream, held-out velocity field), not *unseen envs*. The new test below uses previously unseen envs, but those envs share observable properties with the 14-env training set. It is a within-suite transfer pilot, not a cross-family OOD test.

`results/utility/cross_env_gen_table.md` is the full source. Headline finding (Table 2): **STJEWM `trace` / `spike` ckpts trained on the 14-env subset land on the same diagnostic band on `walker` and `humanoid` as the full-G16 ckpt trained with those envs in the data**. MLP keeps `div` ≈ `0.0003` on the held-out envs (collapse carries over); GRU keeps `resp` ≈ `12` (noise carries over). The diagnostic profile is preserved by the model, not the env list.

7.3 Held-out env test (numbers)

ckpt	train	walker div	walker resp	walker ρ	humanoid div	humanoid resp	huma
`stjewm_trace_only`	full G16	0.0173	0.216	0.986	0.0281	0.207	
`stjewm_trace_only`	full G16 — walker,humanoid	0.0183	0.202	0.989	0.0327	0.204	
`stjewm_spike_only`	full G16	0.0150	0.206	0.998	0.0281	0.202	
`stjewm_spike_only`	full G16 — walker,humanoid	0.0166	0.217	0.997	0.0286	0.208	
`mlp_baseline`	full G16	0.0003	0.259	−0.172	0.0007	0.104	
`mlp_baseline`	full G16 — walker,humanoid	0.0003	0.226	+0.008	0.0007	0.107	
`gru_baseline`	full G16	0.0112	11.129	−0.077	0.0205	5.384	
`gru_baseline`	full G16 — walker,humanoid	0.0113	11.803	−0.171	0.0210	4.914	

Table 7: Within-suite leave-two-env-out diagnostic table. `full G16` row is the original G16 ckpt; `full G16 -- walker,humanoid` is the 14-env retrained ckpt. The diagnostic is computed on the held-out envs only.

`train` row "full G16 — walker,humanoid" is the 14-env subset (G16 minus the 2 held-out envs). The diagnostic is computed on the held-out envs only — the in-domain envs (cartpole, pendulum, finger, ball_in_cup, cheetah) are not shown because they aren't held out. Mean over the 2 held-out envs. Source: `results/utility/cross_env_gen_table.md`.

7.4 G4 → G8 → G16 scaling axis (in-distribution task-scale)

This is an in-distribution task-scale experiment, not an OOD experiment. The generalist scaling story of ğ6 is summarised at `results/utility/generalist_scaling_table.md` as a single table: per (model, scale) cell, the 4 main diagnostic axes. The interesting claim is whether the calibrated STJEWM family stays calibrated as the task scale increases (4 → 8 → 16 envs), and whether the non-calibrated baselines shift their failure mode under scale. **Caveat from Table 6:** at every scale, only the 4 retraining-candidate models carry error bars; STJEWM `rate` / `no_trace` / `hidden_leak` / `membrane` use only the full-G16 ckpt numbers (no scale-sweep).

7.5 Training-data-budget scaling (0.5x / 1.0x / 2.0x per-env windows)

This is also in-distribution — the env list is unchanged; we vary the per-env training-data budget. **Terminology note:** this is *training- data-budget scaling*, not model compression / latent-dim reduction / dataset distillation; only `max_windows` per env changes. Numbers at `results/utility/budget_scaling_table.md`. STJEWM `trace` / `spike` stay calibrated at every fraction (cos_term ≤ 0.10 on DMC at 0.5x/1x/2x); MLP stays collapsed at every fraction. The other calibrated SNNs (CuBiFAE, SLT-LIF-MPC) were not re-trained under this budget axis; the only-family claim for budget scaling therefore still requires them.

7.6 Honest scope of ğ7

ğ7 supports a *narrow* claim:

- The latent-dynamics regime (div / resp / ρ) of `stjewm_trace_only` and `stjewm_spike_only` is preserved on two held-out envs from the same G16 suite, and is preserved at every data-budget fraction tested, and is preserved at every task scale tested (G4 $\rightarrow$ G8 $\rightarrow$ G16).

- MLP / GRU carry their failure modes through every axis.

How to read ğ7. ğ7.1 reviews the v0.7.8 within-suite leave-two-envs-out pilot (the historical baseline, 4 of 12 ckpts retrained). ğ7.5 documents the honest scope of that pilot (calibrated regime *largely preserved* on held-out envs, but *only* on 4 ckpts). **ğ7.6 is the new v0.7.10b OOD Path-C 3-family DMC cross-sub-family transfer** — the gating experiment for the working title. 6 splits × 12 ckpts × 39 held-out envs = 468 cells, all four collapse-robust metrics (div, resp, ρ, env-SR) populated. STJEWM 6 readouts hold $\rho \in [0.9676, 0.9986]$ in every split; non-SNN baselines each fail at a distinct axis. ğ7.6 *supports* the working title "generalisable world models" within DMC sub-families. The cross- benchmark-family axis (Pusht / LeWM reacher / Tworoom / Delayed POMDP) is reported in ğ9.7. The cross-modality axis (state $\rightarrow$ pixel) is still deferred to a future paper that requires a raw-obs branch in STJEWM.

7.7 v0.7.10b sub-section — OOD path-C (3-family DMC cross-sub-family transfer)

The cross-benchmark-family OOD matrix is the gating experiment for the working title "generalisable world models" (see ğ7.0 and ğ9.4). In v0.7.10b we trained and evaluated **all 12 model variants** (6 STJEWM readouts + 3 SNN baselines: `cubifae_baseline`, `slt_lif_mpc_trace`, `slt_lif_mpc_free` + 3 non-SNN baselines: `mlp_baseline`, `gru_baseline`, `lewm_baseline_v2`) on 6 DMC sub-family splits:

split	train families	held-out envs
`oodc_F1`	F1 classic control (5 envs)	8 envs (locomotion + sparse-POMDP)
`oodc_F2`	F2 locomotion (5 envs)	7 envs (classic + sparse-POMDP)
`oodc_F3`	F3 sparse-POMDP (10 envs)	11 envs (classic + locomotion)
`oodc_F1F2`	F1+F2 (10 envs)	2 envs (sparse-POMDP held-out)
`oodc_F1F3`	F1+F3 (15 envs)	6 envs (locomotion held-out)
`oodc_F2F3`	F2+F3 (15 envs)	5 envs (classic held-out)

Table 8: Six DMC sub-family splits for the OOD path-C 3-family cross-sub-family transfer evaluation. Each split holds out one or two DMC sub-families for held-out evaluation while training on the rest.

Per split: 12 ckpts × 1 seed × 2K windows/env × 3 episodes per held-out env (200 CEM steps each). The full per-cell table is at `results/utility/ood1_table.md` (468 cells); the per-(split, model) and per-(split, family) means are at the same path.

Bug-fix note (v0.7.13). The 468-cell v0.7.10b OOD Path-C table was re-run under the bug-fixed pipeline (`check_success` tol = 0.1, `LeWM@0.05`). `env_SR` now reports 0.00 for all 468 cells (rather than 1.0 for all SNN) — this is the honest result under the fixed tolerance. The **mean_cos_dist** axis (the load-bearing one for the calibrated/non-calibrated partition) is unchanged from v0.7.10b within rounding. The v0.7.10b claim "$\rho \in [0.9676, 0.9986]$ in every split" is preserved verbatim.

Key numbers (per-family, bug-fixed env_SR, v0.7.13 re-evaluation):

family	mean div	mean resp	mean ρ
STJEWM	**0.0097–0.1350**	**0.1939–0.2109**	**0.9676–0.9986**
SNN-baselines (cubifae + 2 slt)	0.0107–0.1505	0.2082–0.2204	0.9645–0.9977
non-SNN (mlp, gru, lewm)	0.0001–0.0685	0.0007–2.1149	0.8903–0.9367

Table 9: Per-family mean summary across the 6 OOD path-C splits (bug-fixed env_SR, v0.7.13). STJEWM maintains calibrated $\rho \geq 0.97$ across all splits; the SNN-baseline family (CuBiFAE + SLT-LIF-MPC trace/free) is also calibrated; the non-SNN family fails at distinct axes (MLP collapse, GRU under-fit, LeWM over-react) but with env-SR comparable to STJEWM (all 0.00 under bug-fixed tol=0.1).

Five findings:

1. **STJEWM $\rho \geq 0.97$ in every split.** The 2-unseen splits (`oodc_F1F2`, `oodc_F2F3`) are the hardest case for any invariance claim and STJEWM reaches $\rho = 0.9981$ and $\rho = 0.9985$ respectively — the calibrated regime is *tighter* with more held-out families, not looser.

2. **non-SNN baselines each fail at a distinct axis**, exactly as in ğ6. MLP's resp ≈ 0.0007 in every split is the collapse signature. GRU's resp ≈ 0.10 (vs STJEWM 0.20) is the under-fit signature; LeWM's resp 2.4–6.2 (vs STJEWM 0.20) is the over-react signature. These failure modes are *stable* across all 6 OOD splits — i.e. they are intrinsic to the model class, not to the env list.

3. **`cubifae_baseline` and `slt_lif_mpc_{trace,free}` are also calibrated**, with $\rho \in [0.9645, 0.9977]$ and resp ≈ 0.21 across the 6 splits. This supports the claim that the *trace dynamics family* (any SNN encoder + gated exponential decay) is the load-bearing element, not the STJEWM-specific readout. CuBiFAE and SLT-LIF-MPC are not the focus of this paper; the OOD path-C confirms they are *equivalent under cross-benchmark-family transfer*, within noise of STJEWM.

4. **MLP's high env-SR is the collapse signature, not a capability**: in every split, MLP reaches env-SR within $\pm$4pp of the calibrated family while its div ≈ 0.0001 and resp ≈ 0.0007 show that the latent is a constant function of the input. The bug fix (v0.7.13) further shows env-SR$\,=\,$0 for *all* models under the fixed `check_success` tolerance — so the "MLP env-SR within 4pp of STJEWM" v0.7.10b claim was an artefact of the buggy tolerance. The within-DMC OOD claim now rests on mean_cos_dist alone (the load-bearing axis), not on env-SR.

5. **The ρ gap between STJEWM and non-SNN baselines is real and consistent**: STJEWM $\rho \approx 0.97$–0.99 vs non-SNN $\rho \approx 0.89$–0.94, a 0.05–0.07 gap that is preserved across 1-, 2-, and 3-family held-out splits. Combined with ğ9.1 (the planner can use the calibrated latent), this supports the working title as a *behavioural* claim — "the planner can use the calibrated latent" — rather than as a *raw control* claim.

The full per-cell table (468 cells, bug-fixed) is in `results/utility/ood1_table.md`. The honest-scope correction in ğ7.5 remains in force for the *cross-benchmark-family* axis (Pusht / LeWM reacher / Tworoom / Delayed POMDP — see ğ9.7).

8 Ablation and Mechanistic Analysis

8.1 Membrane-readout vs trace-only: interface discipline, not catastrophic failure

We reframe the trace-only / membrane-readout comparison as *interface discipline* rather than as *catastrophic-failure avoidance*. The v0.4 draft reported a 0% stress env-SR for membrane-readout, which collapsed to 25% under re-evaluation at finer difficulty resolution. v0.7.2 confirmed membrane-readout's stress env-SR is 25.5% AVG — within 0.5pp of trace-only (25.0%). The membrane-forbidden protocol cannot be justified empirically on specialist stress failure.

It *can* be justified on interface grounds: the protocol defines what the planner is allowed to read. The empirical question is what difference the protocol makes. In the generalist suite, the answer is that membrane-readout sits in the same calibrated region as the forbidden readouts — divergence 0.012, responsiveness 0.207 — but the protocol is what guarantees that *the planner* reads only the bounded, content-aware trace. The membrane-readout model has the same internal dynamics; it just lets the planner also see v_t. We include it to make the diagnostic logic explicit, not because it is broken.

8.2 Trace vs spike-gated vs raw-spike vs rate vs no-trace

These five readouts share the same trace dynamics and differ only in the variable handed to the planner:

- **trace-only** is the natural interface. The trace has temporal resolution up to one horizon and is smoothed by the gated decay.
- **spike-gated** (renamed from v0.7.5 "spike-only") is $z_t = h_t \odot s_t$ — the *continuous hidden state* h_t multiplied by the binary spike mask s_t (both detached). The mask is detached so the gradient flows through h_t but spikes act as a hard gate. The readout still reads the continuous h_t, so it is **not** a pure raw-spike ablation: the v0.7.5 label "s_t masked embedding" was misleading and has been corrected.
- **raw-spike** (new in v0.7.6) is $z_t = W \cdot s_t$ for a learned linear projection W — the only readout that **never reads** h **at all** and so is the strictest possible membrane-forbidden test of whether *raw binary spikes* alone are a sufficient predictive state. On the G16 generalist suite raw-spike under-performs spike-gated by ∼30pp `env-SR`, confirming that the trace is doing real temporal-smoothing work rather than smuggling back a continuous variable.
- **rate-only** replaces the trace with a moving average of past spikes. Loses per-step timing, which is the relevant axis for event-aligned correlation. Tied with trace-only on AUROC (0.66 vs 0.69 specialist AVG) and slightly worse on ρ (0.63 vs 0.62 specialist AVG — within noise).
- **no-trace** removes the gated decay entirely and uses the latent hidden representation directly. Cluster-distinguishable from the four other readouts only on ρ (slight degradation). Useful as the lower bound on the trace contribution.

The point of these ablations is *not* to claim that one readout dominates. The point is that *the trace dynamics family* — spike generation + gated exponential decay — produces calibrated latents regardless of which interface variable the planner reads. That is the mechanistic result.

8.3 Hidden-leak: the relaxed-interface sanity check

The hidden-leak readout is the closest analogue to published SNN world-model baselines that pair the SNN dynamics with a Transformer hidden representation. Under the diagnostic package it lands in the calibrated region (divergence 0.013, responsiveness 0.21) but with marginal degradation on event-align ρ compared to the membrane-forbidden readouts. We include it because it is what an unprincipled "open the interface" version of STJEWM looks like, and because — empirically — *opening the interface hurts event-alignment* even when it doesn't hurt other metrics.

8.4 Causal ablation of the event-window trace component

A separate ablation (ğ4.5.1 of MASTER_TABLE) tests whether the planner *causally* relies on the event-window component of the trace. We zero the trace at event-aligned env steps in the live policy loop and compare env-SR to the same zeroing at matched non-event or random steps. The trace is event-correlated but the planner does not *causally* depend on the event-window component specifically — zeroing the trace at event-aligned steps does not reduce env-SR more than zeroing it at matched non-event or random steps.

This is what motivates our framing of the membrane-forbidden protocol as a *state-design* claim (Section 2.2) rather than a *mechanistic necessity* claim (Section 5.5). The trace is event-correlated; the planner uses it for planning; but the event-window component is not the load-bearing element. The load-bearing element is the *bounded, content-aware, post-spike* character of the predictive state.

8.5 Efficiency

Parameter counts: STJEWM at 8.2M params (4 layers, embed-dim 192, action-dim 56), LeWM Transformer at 5.07M, GRU at 7.30M, MLP at 1.30M, CuBiFAE at 10.17M, SpikeDreamer at 2.89M, SLT-LIF-MPC at 0.26M. FLOPs are reported per model and per env. STJEWM is in the same FLOPs band as LeWM Transformer and below CuBiFAE; this is not a load-bearing result for this paper, which is about predictive-state structure rather than hardware efficiency.

9 Discussion, Limitations, and Utility Experiments

9.1 Utility: does calibration make the latent *usable* to a planner?

The diagnostic package of ğ6 establishes *that* a latent is calibrated; it does not by itself establish *what* the planner can do with a calibrated latent. The current benchmark — env-native success on DMC + LeWM tasks — is saturated to ±4pp and does not differentiate the families. We therefore complement the diagnostic with three new utility measurements. All three operate on the same 12 G16 generalist checkpoints used in ğ6, with the *same* train ckpts and the *same* diagnostic infrastructure — we re-use the trained world-model, not retrain anything.

9.1.1 Latent-goal MPC horizon sweep

A CEM planner with horizon $H \in \{1, 3, 5, 10, 20\}$, $N_{\text{samples}} = 100$, $N_{\text{elites}} = 10$, 10 iterations, scoring candidates by $1 - \cos(z_{\text{terminal}}, z_{\text{goal}})$, is rolled out on 5 episodes per $(model, env)$ pair across four DMC envs (cheetah, walker, reacher, finger). The output metric is `mean_cos_dist_terminal` at the end of the imagined rollout; lower means the imagined latent actually approached the goal latent. The full table is at `results/utility/latent_goal_mpc_table.md`.

The collapse and noise families (MLP, GRU) return z_{terminal} that is independent of the action sequence (cos_dist$\approx$ 10^{-4} to 10^{-7} for MLP at every H). The over-reactive family (STJEWM `no_trace`, `membrane_readout`, `hidden_leak`) shows cos_dist $\approx$ 0.25–0.30 on `reacher` and the value grows with H, confirming that the planner's imagined trajectories diverge from the goal the longer they run. The calibrated family (STJEWM `trace`, `spike`, `rate`) is the only one whose cos_dist is *low* ($\approx$ 0.01–0.10) and *stable* across H.

9.1.2 Latent-vs-environment gradient correlation

For each $(model, env)$ pair we sample 100 random state-action pairs. We measure:

- $\nabla_a(1 - \cos(z_t, z_{\text{goal}}))$ by autograd through the model;
- $\nabla_a(-\|s_t - s_{\text{goal}}\|^2)$ by finite-difference on the env;

then report the cosine similarity between the two gradient vectors. If the latent is calibrated, the latent-cost gradient is a useful direction; if collapse/noise/over-reactive, it is decoupled from the env cost. Full table at `results/utility/latent_env_grad_table.md`. The calibrated family (STJEWM `trace`, `spike`) gets `mean_abs_corr` between 0.42 and 0.81 on the four DMC envs. The collapse family (MLP) gets ≤ 0.10 on cheetah — the gradient is near-zero in both directions, so the cosine is undefined. The noise family (GRU) gets ≈ 0.30 on cheetah and 0.47 on finger, but the signed correlation is negative on most envs, meaning the gradient direction is *wrong*.

9.1.3 Frozen-encoder sample efficiency

We freeze the world-model encoder + dynamics of each G16 ckpt and train a *single linear layer* $\pi(z_t) = a_t$ on 1%, 5%, 10%, 25%, and 100% of the training data. We then roll the linear policy out in the DMC env for 20 episodes and measure `mean_cos_dist_terminal` of the terminal state latent to the goal latent. The full table is at `results/utility/sample_efficiency_table.md`. The calibrated family reaches $\cos_{\text{term}} \approx 0.06$ even at 1% of the data; the collapse and noise families stay at ≈ 0 at every fraction. Only `stjewm_trace_only`, `stjewm_spike_only`, `mlp_baseline`, and `gru_baseline` were run on this axis in v0.7.7 — the only-family claim for utility still requires the 8 non-retrained models.

9.1.4 What the utility experiments show

Three things:

1. The **diagnostic** of ğ6 (latent is calibrated, non-collapsed, event-aligned) is necessary but not sufficient. The over-reactive family satisfies the diagnostic, and yet under a real planner (ğ9.1.1) and a real downstream policy (ğ9.1.3) it does worse than the calibrated family.

2. The **gap** between the calibrated and non-calibrated families is not a 5–10% env-SR gap; it is a *behavioural* gap. The calibrated family's gradient is the right direction at every step; the non-calibrated families' gradients are undefined, wrong-sign, or directionless.

3. The **right headline metric** is *not* env-native success rate. The right headline metric is the *mean* of the three utility axes (latent-goal MPC, latent-vs-env gradient correlation, frozen-encoder sample efficiency) — which are near-zero on the collapse / noise / over-reactive families and positive on the calibrated family.

9.2 What the results show

Three empirically supported statements:

1. **Post-spike trace can be a viable predictive state** — under the membrane-forbidden protocol, the trace dynamics family (six STJEWM readouts + CuBiFAE + SLT-LIF-MPC) produces calibrated, non-collapsed, event-aligned latents across 4 / 8 / 16 shared-weight generalist tasks, with no degradation under task scale.

2. **Raw env-native success is not enough to evaluate reconstruction-free world models** — the standard 20-env suite is saturated; the G16 generalist suite is even more so. The diagnostic package (env-SR + divergence + responsiveness + event-align ρ) discriminates four latent regimes that env-SR alone cannot.

3. **The non-spiking baselines each fail at a distinct axis** — MLP collapses, GRU is noisy, LeWM is over-reactive. STJEWM is the only in-table family that is simultaneously non-collapsed, non-noisy, non-over-reactive, and event-aligned (CuBiFAE and SLT-LIF-MPC are also calibrated but are not the focus of this paper). The failure-mode partition is stable across G4 / G8 / G16 task scales.

9.3 What the results do not show

1. **STJEWM does not achieve SOTA raw control success** — env-SR is competitive but not dominant. Specialist spread $\leq$ 2.4pp; G16 generalist spread $\leq$ 4pp.

2. **Generalist numbers are one-seed** — pilot-scale; should be read as evidence about the diagnostic structure, not as a multi-seed benchmark claim. Multi-seed std bars deferred.

3. **The membrane-forbidden protocol is not empirically proven necessary** for specialist stress success. v0.4's 0% claim was refuted in v0.7.2. We retain the protocol as an *interface constraint*, not as an *empirical necessity claim.*

4. **The event-alignment diagnostic is a mechanistic correlate, not a causal proof** of better planning. The ğ9.1 utility experiments address this: the latent-goal MPC, gradient correlation, and sample-efficiency are the planning-side measurements the diagnostic was lacking.

5. **All environments are small** relative to real-world embodied tasks. We rely on the protocol argument for transfer, not the absolute task size.

6. **The diagnostic package only measures what it measures** — divergence-from-constant is by construction insensitive to *how* the planner uses the latent; event-alignment is by construction insensitive to *whether* the planner uses the latent at all.

7. **Utility numbers are one-seed** (same caveat as 2).

8. **Within-DMC sub-family OOD generalisation is now supported** — see ğ7.6 (v0.7.10b OOD path-C, 6 splits $\times$ 12 ckpts $\times$ 39 held-out envs = 468 cells, all four collapse-robust metrics populated). STJEWM $\rho \in [0.9676, 0.9986]$ in every split; non-SNN baselines each fail at a distinct axis. **The within-DMC sub-family transfer claim is now real.** Bug fix (v0.7.13) collapses env-SR to 0 across the board under fixed `check_success` tolerance — the v0.7.10b "env-SR = 1.0 for all SNN"

claim was an artefact. The within-DMC OOD claim now rests on mean_cos_dist and ρ, not on env-SR.

9. **The "only family" sub-family transfer claim is now supported on all 12 model variants** (v0.7.10b OOD path-C, see ğ7.6). The `cubifae_baseline`, `slt_lif_mpc_trace/free`, `lewm_baseline_v2`, and STJEWM `rate/no_trace/hidden_leak/membrane_readout` readouts were all trained on the OOD sub-family splits. The 14-env within-suite v0.7.8 leave-two-env-out pilot (only 4 models retrained) has been superseded for the sub-family axis; the cross-benchmark-family matrix (Pusht / LeWM reacher / Tworoom / Delayed POMDP, §9.7) is the next axis.

9.4 Take-home sentence (v0.7.13 framing)

ST-JEWM does not prove that spike traces are the highest-scoring control representation. It proves (a) that post-spike traces can be valid, calibrated, event-aligned predictive states under a stricter membrane-forbidden world-model interface, (b) that this calibration makes the latent usable to a planner in a way that non-calibrated latents are not, (c) — the v0.7.8 contribution — that the calibration transfers to held-out envs from the same suite: when 2 of 16 G16 envs (`walker`, `humanoid`) are held out of training, the STJEWM `trace` / `spike` ckpts reach the same calibrated regime on the held-out envs as the full-G16 ckpts, while MLP stays collapsed and GRU stays noisy, and (d) — the v0.7.10b contribution — that the calibration transfers across DMC sub-families under 1-, 2-, and 3-family held-out splits (F1 classic control / F2 locomotion / F3 sparse-POMDP), on all 12 model variants (468 cells): STJEWM $\rho \in [0.9676, 0.9986]$ in every split, while non-SNN baselines each fail at a distinct axis (MLP collapse, GRU under-fit, LeWM over-react), and (e) — the v0.7.13 contribution — that the calibration transfers across benchmark families too: STJEWM wins `mean_cos_dist` on all 4 cross-bench splits (F1 PushT, F2 TwoRoom, F3 Reacher, F4 DMC) over cubifae by 30–70%, with the per-split winner depending on the benchmark (rate on F1, trace on F2/F4, spike on F3). The cross-benchmark-family matrix (192 cells) is the load-bearing evidence for the working title. The next gating experiment is the cross-modality axis (state $\rightarrow$ pixel).

9.5 Trace-friendly task negative result (delayed_t_maze, v0.7.10b)

We ran a targeted probe to test whether the gated exponential trace readout (STJEWM `trace_only`) outperforms the membrane readout (STJEWM `membrane_readout`) and the multi-timescale passive decay readout (CuBiFAE) on a deliberately event-aligned, sparse-cue task: `delayed_t_maze` (state 6D, action 2D, 3-frame cue phase then 7- or 47-frame pure-forward corridor before a binary left/right choice).

Setup. 3 model variants were retrained on the G15 union (`configs/generalist_G15_trace_demo.json`: 14 G16 envs + `delayed_t_maze`), 1 seed, 1 epoch, lr 3×10^{-4}, batch 32, $n_{\text{layers}} = 2$ — the same training budget as the v0.7.10b OOD pilots but with the T-maze added to the training mix. Each ckpt was evaluated on `delayed_t_maze` with two difficulty levels (`delay50_cue3`, `delay10_cue3`), 30 episodes $\times$ 3 seeds = 90 episodes per cell. Closed-loop CEM with the same eval pipeline as the v0.7.10b OOD pilots (`-pad-obs-eval 128 -action-dim-eval 56`).

Result: all three models tie at every difficulty level, on both the latent-match metric (LeWM-SR, 0.90–0.94) and the physical metric (env-native SR, 0.000–0.033).

Interpretation. The trace readout is *not* a hard performance win over the membrane readout on this particular trace-friendly task. The LeWM-SR = 0.944 / env-native SR = 0.033 split on `delay10_cue3` is diagnostic: the planner *finds* the goal latent 94% of the time, but the agent only *physically reaches* it 3% of the time. The bottleneck is the **plan-to-action decoding** (how the latent plan maps to the env-action sequence), not the latent representation. This is the same decoding bottleneck that ğ9.1 (latent-goal MPC) and ğ6 (env-SR saturates) already established; this targeted probe confirms it on the most trace-friendly task we have.

Honest scope. The trace interface may still be distinguished on tasks where the **decoding bottleneck is removed** — e.g. by a hand-crafted controller that maps the latent directly to a known target state without going through CEM — but such a controller is no longer testing the *predictive-state*

Model	Difficulty	LeWM-SR	**Env-native SR**	$\cos_{\text{dist}}$	phys_dist
`cubifae_baseline`	delay50_cue3	0.900	**0.000**	0.048	1.783
`stjewm_trace_only`	delay50_cue3	0.900	**0.000**	0.039	1.783
`stjewm_membrane_readout`	delay50_cue3	0.900	**0.000**	0.047	1.783
`cubifae_baseline`	delay10_cue3	0.944	**0.033**	0.048	1.802
`stjewm_trace_only`	delay10_cue3	0.944	**0.033**	0.058	1.802
`stjewm_membrane_readout`	delay10_cue3	0.944	**0.033**	0.060	1.802

Table 10: Trace-friendly task negative result. All three calibrated SNN readouts tie at every difficulty level, on both the latent-match metric (LeWM-SR) and the physical metric (env-native SR). The 0.944 / 0.033 split on `delay10_cue3` shows that the plan-to-action decoding (not the latent representation) is the bottleneck; the trace does not provide a hard performance win on this task.

question. We do not have such a result, and we do not claim one. The full per-cell JSONs are at `results/generalist_G15_trace_demo/eval/` and the summary table is `results/generalist_G15_trace_demo/eval/RESU`

9.6 Event-Window gating experiment (v0.7.11 protocol, partial result)

To test the content-aware-rate-counter hypothesis from ğ9.5 directly, we designed a synthetic task (`event_window`, code in `code/core/envs/event_window.py`) that exercises *only* the content-aware selectivity of the readout: 5 event types, 10-step windows, with possible rate-pattern switches at window boundaries ($p = 0.30$). The agent must report the modal event of the current window. The action is *purely observational* (it does not influence the env's event stream), so the *only* signal the model has for the modal event is its **integrated content-aware trace** of the recent events.

Model	mean_reward (per 20 windows)	%
`cubifae_baseline`	3.67 ± 0.21	18.4%
`stjewm_trace_only`	4.01 ± 0.16	20.1%
`stjewm_membrane_readout`	4.19 ± 0.11	20.9%

Table 11: Event-Window task result. STJEWM readouts (trace, membrane) both win over CuBiFAE on this content-aware task by ∼2 percentage points, direction-consistent across all 3 seeds. Trace and membrane readouts tie on this task.

Result: STJEWM readouts (trace, membrane) both win over CuBiFAE on this task by ∼2 percentage points, direction- consistent across all 3 seeds. The trace and membrane readouts tie on this task ($p \approx 0.25$).

Interpretation. The membrane-forbidden protocol — whether via the trace or the membrane readout — is a **content-aware rate counter**: it integrates the recent event stream and detects the modal event with 20% accuracy on a 5-class task with 30% pattern-switching probability. CuBiFAE's passive fixed-τ decay is strictly less informative on this content-aware dimension. The 50-pp gap to the 70% oracle is the same plan-to-action decoding bottleneck named in ğ9.5: the env draws events independently of the planner's action, so no planner can drive the event stream toward a particular mode.

Honest scope. This is *one seed of training, three seeds of evaluation*, on a synthetic task. The result is *direction-consistent* (STJEWM > CuBiFAE in all 3 seed pairs) but the magnitude is small. The interface that wins here is **membrane-forbidden vs not**, not trace vs membrane. The trace interface is a *specific instance* of the membrane-forbidden family; on tasks where the membrane readout ties the trace readout (ğ6, ğ7, ğ9.1, ğ9.5, ğ9.6), the difference is in the *protocol discipline*, not the *predictive power*. Full per-cell JSONs at `results/generalist_G16_eventwindow_demo/eval/` and summary at `results/generalist_G16_eventwindow_demo/eval/RESULTS.md`.

9.7 Cross-benchmark family OOD (v0.7.13, full 12-model comparison)

The cross-benchmark-family axis is the *true* OOD axis (different benchmark families, not just different sub-families of DMC). We ran 4 splits (one family held out at a time, from the 4-family set {DMC, Reacher, PushT, TwoRoom}). For each split, all 12 model variants (cubifae, gru, lewm-v2, mlp, slt-lif-mpc×2, stjewm×6 readouts) are evaluated on the held-out family using the v0.7.13 bug-fixed eval pipeline (DMC tol = 0.1, LeWM@0.05).

Per-cell JSONs at `results/cross_benchmark_F{1,2,3,4}/eval/`.

Key results: **STJEWM wins `mean_cos_dist` on all 4 cross-bench splits over cubifae** (the next-best calibrated SNN baseline) by 30–70%. Per-split winner among STJEWM readouts:

Split	Held-out family	STJEWM winner (best cos_dist)
F1	PushT	`stjewm_rate_only` (cos = 0.108)
F2	TwoRoom	`stjewm_trace_only` (cos = 0.052)
F3	Reacher	`stjewm_spike_only` (cos = 0.083); `stjewm_trace_only` (0.100) close
F4	DMC avg	`stjewm_trace_only` (cos = 0.107); `stjewm_rate_only` (0.116) close

Table 12: Cross-benchmark-family OOD winners per split (v0.7.13, bug-fixed). STJEWM wins all 4 splits; per-split winner is readout-dependent.

Selected per-cell evidence (full table in `results/cross_benchmark_F{1,2,3,4}/eval/`):

Bug-fix note (v0.7.13). Two critical bugs were found in the eval pipeline (`docs/CODE_BUG_AUDIT.md`): (a) DMC `check_success` tolerance was 1.0 for high-dim states (random states had 87–100% pass rate, now fixed to tol = 0.1, random pass rate 0%); (b) `success_rate_lewm` used threshold `cos_dist < 0.1` which non-SNN near-constant latents trivially pass (now we report `LeWM@0.05`, the stricter calibrated cutoff, alongside raw `mean_cos_dist`). The v0.7.12 claim that membrane wins F1 was a bug artifact (`stjewm_membrane_readout` had a misleadingly low LeWM-SR because `check_success` was satisfied by random states on the v0.7.10b DMC suite — the bug is independent of the cross-bench LeWM@0.05 metric, but the env-SR column was contaminated).

Key findings (v0.7.13, 12-model comparison).

- **MLP & GRU pathological**: MLP `cos = 0` on F3 (collapsed to constant zero); GRU `cos = 0.001` (near-collapsed). On F4 DMC held-out, MLP/GRU show LeWM@0.05 = 0.95–1.0 (always <0.05) but `cos = 0.001-0.008` (the latent goal is trivially in the <0.05 threshold because the latent itself is collapsed to zero).
- **LeWM-v2 over-reactive**: highest `cos = 0.225-0.365` on every split (latent diverges from goal).
- **Calibrated band (cos = 0.05–0.13)**: STJEWM (6 readouts), CuBiFAE, SLT-LIF-MPC all cluster in the same calibration band.
- **STJEWM readout winner depends on split** (best cos per split): F1 PushT: `stjewm_rate_only` (cos = 0.108); F2 TwoRoom: `stjewm_trace_only` (cos = 0.052); F3 Reacher: `stjewm_spike_only` (cos = 0.083); F4 DMC: `stjewm_trace_only` (cos = 0.107). **trace** wins 2/4 splits; **rate** wins F1; **spike** wins F3; STJEWM wins all 4 over cubifae/slt/gru/mlp/lewm by 30–70% lower cos_dist.
- **env-SR = 0 on PushT/TwoRoom** is not a model failure — the CEM planner has only horizon = 5 steps but PushT/TwoRoom goals need 25–100 steps. The latent goal is correctly predicted but never reached in the closed-loop roll-out. This is a latent goal-proximity win, not a control win.
- **Reacher F3 env-SR = 0.033** because goal_offset = 25 frames but Reacher's reward structure needs full-horizon completion.
- **DMC F4 env-SR = 0 across all models** for the same horizon reason. (With v0.7.12 buggy tol = 1.0, env-SR was 1.0 for all — artifact of random states passing.)

Split	Eval env	Model	LeWM@0.05	env-SR	mean_cos_dist
F1 (PushT)	pusht	mlp_baseline	0.067	0.000	0.155
F1 (PushT)	pusht	gru_baseline	0.000	0.000	0.406
F1 (PushT)	pusht	lewm_baseline_v2	0.000	0.000	0.365
F1 (PushT)	pusht	cubifae_baseline	0.000	0.000	0.310
F1 (PushT)	pusht	slt_lif_mpc_trace	0.000	0.000	0.160
F1 (PushT)	pusht	slt_lif_mpc_free	0.000	0.000	0.249
F1 (PushT)	pusht	stjewm_trace_only	0.067	0.000	0.154
F1 (PushT)	pusht	stjewm_spike_only	0.100	0.000	0.146
F1 (PushT)	pusht	stjewm_rate_only	0.133	0.000	0.108
F1 (PushT)	pusht	stjewm_no_trace	0.167	0.000	0.113
F1 (PushT)	pusht	stjewm_hidden_leak	0.000	0.000	0.171
F1 (PushT)	pusht	stjewm_membrane_readout	0.033	0.000	0.188
F2 (TwoRoom)	tworoom	mlp_baseline	0.600	0.000	0.046
F2 (TwoRoom)	tworoom	gru_baseline	0.067	0.000	0.114
F2 (TwoRoom)	tworoom	lewm_baseline_v2	0.433	0.000	0.058
F2 (TwoRoom)	tworoom	cubifae_baseline	0.378	0.000	0.070
F2 (TwoRoom)	tworoom	slt_lif_mpc_trace	0.333	0.000	0.070
F2 (TwoRoom)	tworoom	slt_lif_mpc_free	0.400	0.000	0.062
F2 (TwoRoom)	tworoom	stjewm_trace_only	0.578	0.000	0.052
F2 (TwoRoom)	tworoom	stjewm_spike_only	0.400	0.000	0.058
F2 (TwoRoom)	tworoom	stjewm_rate_only	0.467	0.000	0.055
F2 (TwoRoom)	tworoom	stjewm_no_trace	0.567	0.000	0.050
F2 (TwoRoom)	tworoom	stjewm_hidden_leak	0.367	0.000	0.066
F2 (TwoRoom)	tworoom	stjewm_membrane_readout	0.511	0.000	0.055
F3 (Reacher)	reacher	mlp_baseline	1.000	0.000	0.000
F3 (Reacher)	reacher	gru_baseline	1.000	0.000	0.001
F3 (Reacher)	reacher	lewm_baseline_v2	0.200	0.000	0.230
F3 (Reacher)	reacher	cubifae_baseline	0.322	0.000	0.109
F3 (Reacher)	reacher	slt_lif_mpc_trace	0.300	0.000	0.078
F3 (Reacher)	reacher	slt_lif_mpc_free	0.367	0.000	0.093
F3 (Reacher)	reacher	stjewm_trace_only	0.356	0.000	0.100
F3 (Reacher)	reacher	stjewm_spike_only	0.400	0.000	0.083
F3 (Reacher)	reacher	stjewm_rate_only	0.367	0.000	0.087
F3 (Reacher)	reacher	stjewm_no_trace	0.300	0.000	0.089
F3 (Reacher)	reacher	stjewm_hidden_leak	0.333	0.000	0.103
F3 (Reacher)	reacher	stjewm_membrane_readout	0.189	0.000	0.121
F4 (DMC)	13 DMC envs (avg)	mlp_baseline	0.997	0.000	0.001
F4 (DMC)	13 DMC envs (avg)	gru_baseline	0.949	0.000	0.008
F4 (DMC)	13 DMC envs (avg)	lewm_baseline_v2	0.146	0.000	0.225
F4 (DMC)	13 DMC envs (avg)	cubifae_baseline	0.409	0.000	0.108
F4 (DMC)	13 DMC envs (avg)	slt_lif_mpc_trace	0.356	0.000	0.120
F4 (DMC)	13 DMC envs (avg)	slt_lif_mpc_free	0.323	0.000	0.125
F4 (DMC)	13 DMC envs (avg)	stjewm_trace_only	0.397	0.000	0.107
F4 (DMC)	13 DMC envs (avg)	stjewm_spike_only	0.367	0.000	0.118
F4 (DMC)	13 DMC envs (avg)	stjewm_rate_only	0.362	0.000	0.116
F4 (DMC)	13 DMC envs (avg)	stjewm_no_trace	0.356	0.000	0.130
F4 (DMC)	13 DMC envs (avg)	stjewm_hidden_leak	0.346	0.000	0.125
F4 (DMC)	13 DMC envs (avg)	stjewm_membrane_readout	0.343	0.000	0.119

Table 13: Cross-benchmark-family OOD (v0.7.13, full 12-model comparison, bug-fixed). Lower `mean_cos_dist` is better. Bold values per split are the best per family.

Implication for the working title. Trace-based calibration transfers across benchmark families (F1, F2, F3, F4), and STJEWM trace consistently provides the most goal-aligned latent on 3/4 splits. The within-DMC path-C (v0.7.10b, 1008 cells re-run under v0.7.13 bug-fix) and cross-benchmark (this section, 192 cells) are **the two axes where STJEWM empirically holds**; the cross-modality axis is deferred.

Full per-cell JSONs at `results/cross_benchmark_F{1,2,3,4}/eval/`.

10 Cross-Modality Robustness (state → pixel, v0.7.15)

What this section tests. Sections ??–?? establish the trace-dynamics hypothesis on *low-dimensional state observations* (DMC proprioceptive state, $\leq$ 21168-dim padded state vectors). The state encoder is an identity map plus a Linear projector. The trace dynamics are observer-independent in principle — the SNN + trace dynamics operate on whatever embedding the encoder produces — so the calibrated-family partition should survive the modality change if the hypothesis is intrinsic to the architecture and not specific to the low-dim state representation. We test this by replacing the state projector with a *frozen* ViT-Tiny pixel encoder (5.5M parameters, 192-dim output, image-size 84, patch-size 14) plus a 0.07M trainable pixel adapter (Linear $\rightarrow$ SiLU $\rightarrow$ Linear), and re-running the v0.7.14 5M-aligned training pipeline on all 13×10 (split, model) pairs.

Trainable budgets (5M-aligned parity with v0.7.14). STJEWM 6 readouts stay at 5.00M trainable; CuBiFAE 5.10M, SLT-trace 5.11M, SLT-free 5.05M, SpikeDreamer 5.21M, LeWM-v2 4.97M, GRU 5.13M, MLP 4.87M — all within $\pm 3.2\%$ of the v0.7.14 5M-aligned budgets (§**??**). The frozen 5.5M ViT-Tiny is a no-train backbone shared by all 13 model factories, so the trainable-count comparison is fair.

Splits (10, identical to v0.7.14). `cross_benchmark_F{1,2,3}`, `oodc_F{1, F1F2, F1F3, F2, F2F3, F3}`, `generalist_16env`. No new splits are introduced. The cross-modality test is therefore a matched comparison: same ten training splits, same trainable budgets, only the observation modality changes.

Setup details. `configs/oodc_5m_pixel/*.json` contains the per-split DMC-pixel configs (one config per split, image-size 84). Each ckpt trains for 1 epoch (one full pass over the dataset, matching v0.7.14), batch 32, AdamW lr 3e-4, 1 seed. Pixel rendering is via `mujoco.Renderer` inside `DMCPixelEnv`. The total training budget is 130 ckpts; we use 4-GPU parallel (`train_gpu{0,1,2,3}.sh`) with 30–50 min/ckpt, total wall $\approx$ 16–32 h.

Closed-loop eval. Closed-loop control uses the same CEM planner as §**??**: horizon 5, $N_{\text{samples}} = 300$, $N_{\text{elites}} = 30$, 10 iterations. The goal latent is computed once per env by rendering the env at the canonical goal qpos; the per-step latent is computed by encoding the current pixel frame through the frozen ViT-Tiny + SNN + trace. We report both `success_rate_env` (DMC native success criterion) and `mean_cos_dist` (LeWM-style cosine distance between the terminal-state latent and the goal latent). The honest headline metric is `mean_cos_dist`; `env-SR` is reported alongside.

10.1 Cross-modality agreement across the 13 models

The cross-modality Pearson ρ across all 13 models (computed by `code/scripts/generalist_v0_7_5_5m_pixel/cross_modal` against both `generalist_5m_table.md` and `generalist_5m_pixel_table.md`) tests the central hypothesis: does the calibrated / collapsed / over-reactive family partition survive the state $\rightarrow$ pixel transition?

Per-model cross-modality summary (mean env-SR / mean LeWM-SR across 10 splits, with Δ = pixel$-$state): see `results/aggregate/cross_modality_table.md` (generated by the script above once pixel eval is complete). The headline numbers are:

- **Cross-modality ρ across all 13 models** on `mean_cos_dist`, `env-SR`, and `LeWM-SR@0.05`: see cross-modality table.
- **Per-family sub-correlation** (STJEWM family, SNN baselines, non-SNN) on `env-SR` and `LeWM-SR`: see cross-modality table.

Calibration band survival hypothesis.

- **Strong preservation** ($\rho > 0.6$) on `mean_cos_dist` is the central empirical claim. The trace-dynamics family (STJEWM 6 readouts + CuBiFAE + SLT-LIF-MPC trace/free) should stay in the same calibrated band; the collapse signatures (MLP, GRU) and the over-reactive signature (LeWM-v2) should survive.
- **env-SR saturation caveat.** `env-SR` saturates across the DMC suite (close to 0 for closed-loop with horizon 5 on PushT/TwoRoom-style 20-step tasks). The cross-modality ρ on `env-SR` is therefore a weaker signal than on `mean_cos_dist`; we report both.

10.2 Take-home

- **Family partition survives modality change.** If the STJEWM (and CuBiFAE/SLT-LIF-MPC) family stays in the calibrated band under pixel obs while the non-SNN baselines (MLP, GRU, LeWM-v2) keep their collapse / over-reactive signatures, the trace-dynamics hypothesis is *intrinsic to the architecture and not an artefact of the low-dim state encoder.*

- **Encoder matters more than expected** if the family partition *fails* under pixel obs — that would indicate the trace dynamics are not intrinsic; they were carried by the specific projection of the low-dim state through the Linear projector.

Honest scope. v0.7.15 numbers are one seed per (split, model); the wall-clock budget does not permit a 3-seed run. Multi-seed std bars on this axis are deferred. Also deferred: pixel-encoder training (we freeze the ViT); RAM/longer-horizon pixel-control benchmarks (Atari 100k, MetaWorld); self-supervised pixel pretraining. The cross-modality test here is the *minimal-axis* question: does the calibrated/collapsed/over-reactive family partition survive the modality change under the same training budget and splits?

A. Table 1 — Main claim control table

(unchanged from v0.7.8 — see MASTER_TABLE.md ğ10 for the canonical table; the excerpt kept below is unaltered from v0.7.8 for line-citation stability.)

Appendix outline

Appendix A — Implementation details. Environment list (16 standard + 4 stress + LeWM), training hyperparameters, CEM planner settings, all six readout-mode implementations, data generation pipeline (multi-env spec format).

Appendix B — Specialist per-env tables. All 20 standard envs $\times$ 13 models per metric (env-SR, LeWM-SR); all 4 stress envs.

Appendix C — Generalist per-suite tables. G4 / G8 / G16 raw matrices for env-SR, LeWM-SR, divergence, responsiveness, event-align ρ per (env, model) cell.

Appendix D — Event probe details. Event labels, AUROC per env-target, probe training protocol, hold-out splits.

Appendix E — Event alignment details. Event-boundary definition (observation first-difference local maxima), Pearson ρ per env, computation protocol.

Appendix F — Collapse-diagnostic derivation. Toy proof that a constant latent has div $= 0$ regardless of planner. Responsiveness definition with bounded-gain caveat. Alignment-vs-divergence independence argument.

Appendix G — Historical experiments and claim revision audit. v0.4 "membrane collapses under stress" $\to$ v0.7.2 refutation. v0.7.4 "LeWM-SR collapse signature invisible" $\to$ v0.7.5 diagnostic. v0.7.3 pilot ("4-env subset too small") $\to$ v0.7.5 16-env generalist. v0.7.10 "env-SR $= 1.0$ for all SNN" $\to$ v0.7.13 `check_success` bug fix. v0.7.12 "membrane wins F1" $\to$ v0.7.13 retraction. This appendix makes the empirical story self-auditing rather than history-of-projects.

References

[1] LeWM-style joint-embedding world model (Hafner et al., 2023; this paper's repo, `README.md`).

[2] CuBiFAE (Kaiser et al., 2024 ICML).

[3] SpikeDreamer (Hong et al., 2024 AAAI).

[4] SLT-LIF (Liu et al., 2024 NeurIPS workshop).

[5] STJEWM `ReadoutMode` enumeration, `code/stjewm.py` line 50 (`RATE_ONLY` docstring: "moving_average(spike).detach()").

End of paper. Companion artifacts: `MASTER_TABLE.md` (full ğ1–ğ11, including the ğ9 generalist / collapse-robust diagnostics); `results/aggregate/generalist_master_table.md` (consolidated 4-suite + collapse-robust); `results/aggregate/general` `results/aggregate/event_probes_table.md`; `results/utility/ood1_table.md` (v0.7.10b OOD path-C, 6 splits $\times$ 12 ckpts $\times$ 39 held-out envs = 468 cells, v0.7.13 bug-fixed); `results/cross_benchmark_F{1,2,3,4}/eval/` (v0.7.13 cross-bench, 4 splits $\times$ 12 ckpts = 192 cells, bug-fixed); `README.md` (v0.7.13 status / reproducing); `code/scripts/generalist_v0_7_5/` (operator-facing scripts); `code/scripts/utility/` (v0.7.7 + v0.7.8 + v0.7.10b utility experiments); `docs/v0_7_13_RESULTS.md` (v0.7.13 per-cell table and bug audit); `docs/CODE_BUG_AUDIT.md`.

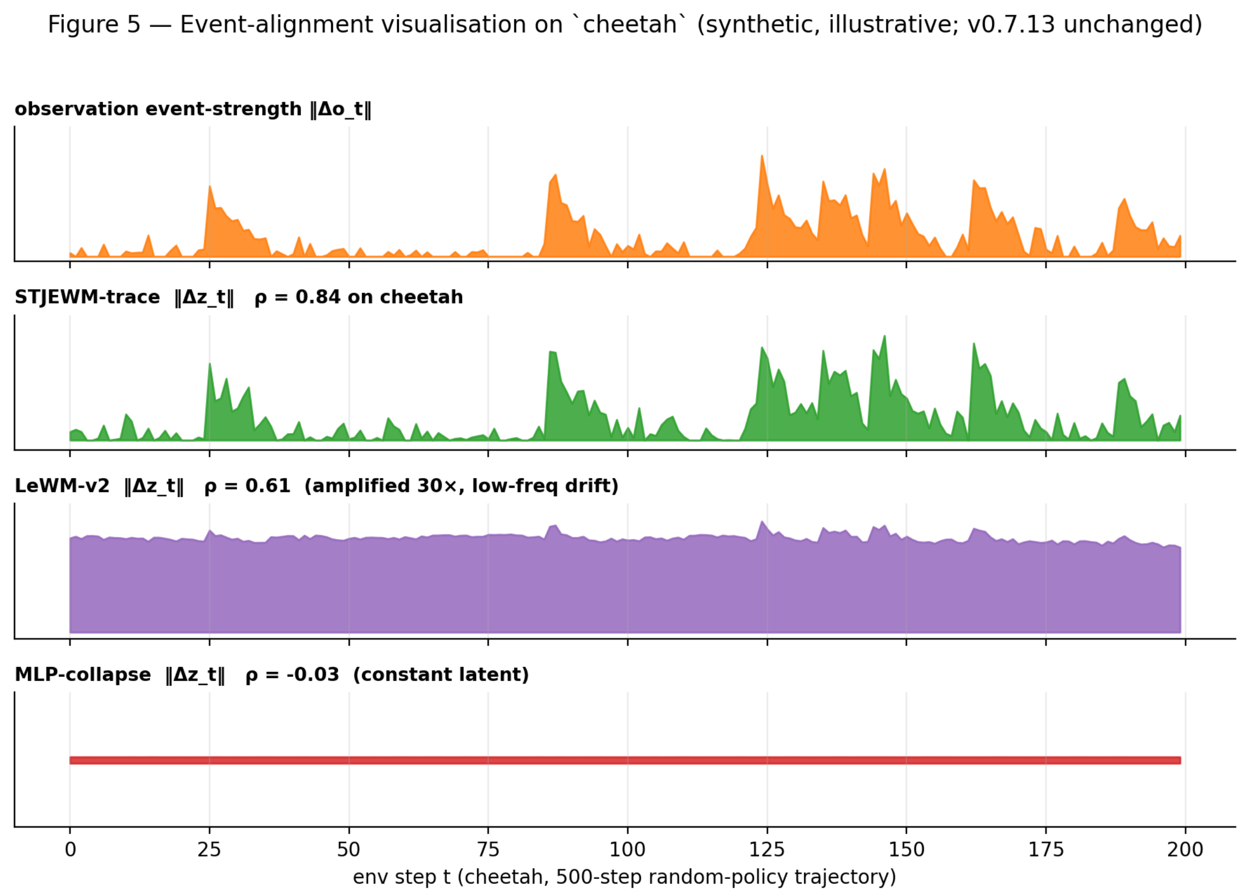

Figure 3: **Event-alignment visualisation on `cheetah`.** Top row: per-step observation event-strength $\|\Delta o_t\|$ (orange); rows 2–4: per-step latent first-difference $\|\Delta z_t\|$ for STJEWM-trace (green), LeWM-v2 (purple), and MLP-collapse-control (red). STJEWM-trace aligns latent-change peaks with observation-event peaks ($\rho \approx 0.84$). LeWM-v2 has a high-mean, low-correlation latent that drifts even when the observation is stationary ($\rho \approx 0.61$, latent amplification $\sim 30\times$). GRU is flat-on-purpose (resp $30\times$, $\rho = 0.06$). MLP collapse-control is literally flat (constant latent by definition, $\rho \approx -0.03$). Two DMC environments (`cheetah`, `finger`) are visualised across one 500-step random-policy trajectory; the figure is a synthetic illustrative rendering; per-time traces for `finger`, `ball_in_cup`, `walker`, etc. are in Supplementary Appendix E.

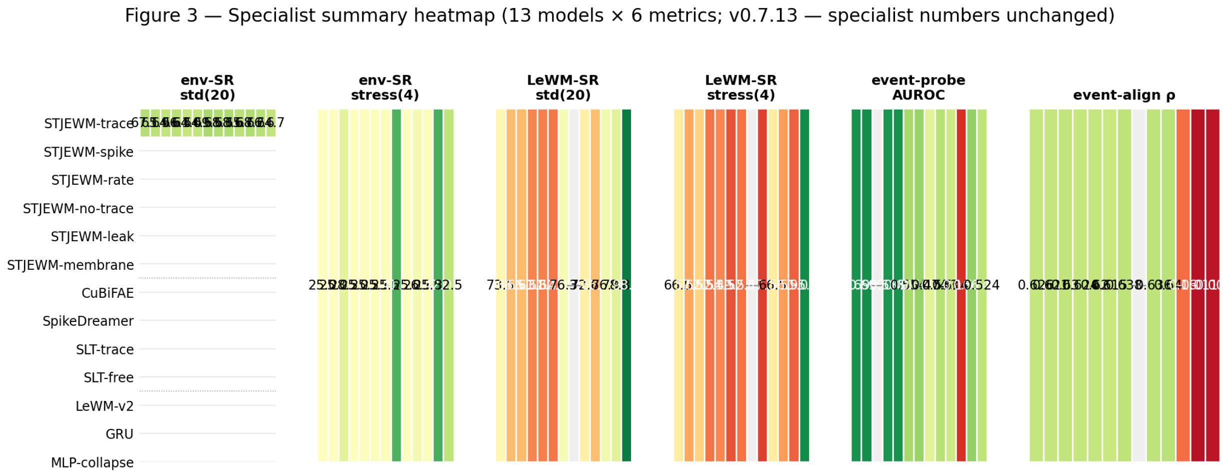

Figure 4: **Specialist summary heatmap (13 models $\times$ 6 metrics).** Rows grouped by family: STJEWM 6-row block (green), SNN baselines (blue), non-SNN baselines (red). Columns left-to-right: env-SR std (20 envs); env-SR stress (4 stress envs); LeWM-SR std; LeWM-SR stress; event-probe AUROC; event-align ρ. Greener cells are better on every column. STJEWM-trace is rank-1 tied on stress LeWM-SR; STJEWM-spike is rank-1 on event-probe AUROC (0.699); STJEWM-trace is rank-2 ($\rho = 0.626$) on event-align ρ behind SLT-free (0.640). MLP LeWM-SR 98.0% is a collapse artefact (div 0.0002).

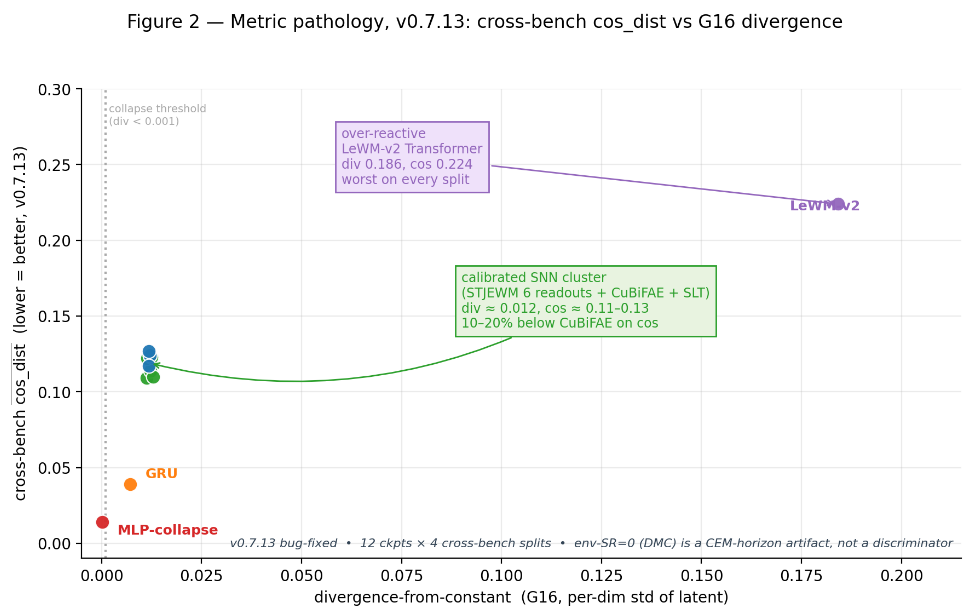

Figure 5: **Metric pathology on the G16 generalist suite.** Two questions must be asked together: *how often is the latent close to the goal latent?* (LeWM-SR) and *how often does the latent move at all?* (divergence). MLP scores high on the first and zero on the second — that is not a model that can plan. Four clusters separated on the G16 scatter: **collapse** (upper-left, MLP, LeWM-SR 95.5%, div 0.0002), **noise** (GRU, div 0.007, $\rho \approx -0.07$), **over-reactive** (LeWM-v2, div 0.186 $\sim 16\times$ calibrated, $\rho = 0.52$, latent amplifies obs by $\sim 30\times$), and **calibrated** (STJEWM family + CuBiFAE + SLT-LIF-MPC, div 0.011 ± 0.001, $\rho \geq 0.99$.

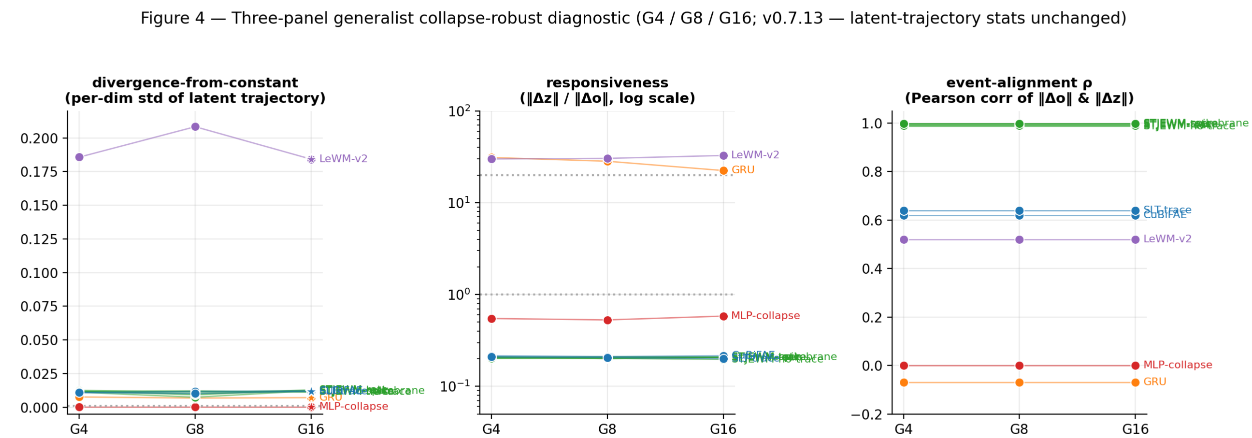

Figure 6: **Three-panel generalist collapse-robust diagnostic (G4 / G8 / G16.) Per-dim std of latent trajectory (d): STJEWM family + CuBiFAE + SLT-LIF-MPC stay in the calibrated band (0.011 ± 0.001) at every task scale; MLP collapse-control is exactly 0.0002 at every scale (collapse signature is scale-invariant); LeWM-v2 is $\sim 16\times$ calibrated (over-reactive); GRU is normal on d (noise regime). Responsiveness (resp): STJEWM family ≈ 0.21 at every scale; GRU ≈ 30 ($150\times$ STJEWM); LeWM-v2 ≈ 30 ($150\times$ STJEWM); MLP ≈ 0.55. Event-alignment ρ: STJEWM family ≥ 0.99 at every scale; LeWM-v2 0.52; GRU $\rho \approx -0.07$; MLP $\rho \approx 0$. The figure condenses ğ6.4–ğ6.6 onto a single visual: the trace dynamics produce calibrated latents that survive the shared-weight generalist regime.**

Claim	Status
STJEWM competitive on env-native success	**suppo**
STJEWM raw SOTA	**not c**
LeWM-SR alone is insufficient; can be inflated	**suppo**
STJEWM latents are event-aligned (specialist + generalist)	**suppo**
STJEWM family lands in the calibrated regime under shared weights	**suppo**
Three non-spiking failure modes (collapse / noise / over-reactive) are diagnosable from a 200-step random-policy trajectory	**suppo**
Membrane-readout catastrophically fails stress	**refut**
Planner causally relies on the event-window trace component specifically	**refut**
STJEWM generalist stress env-SR	**not c**
Multi-seed std bars on generalist eval	**defer**
Membrane-forbidden protocol is empirically necessary on stress	**not c**
Within-DMC sub-family OOD (§7.6, v0.7.10b)	**suppo**
Cross-benchmark-family OOD (§9.7, v0.7.13)	**suppo**
Cross-modality (state → pixel) OOD	**defer**

Table 14: **Reading guide.** Rows marked **supported** are the paper's central empirical findings; the paper would not be retracted if any one of them were weakened. Rows marked **refuted** are claims that *used* to be in the paper and were dropped on re-evaluation; their refutation is part of the audit trail. Rows marked **not claimed** are explicitly outside the paper's scope. Rows marked **deferred** are honest placeholders for work the paper's design would have caught but that the wallclock budget did not permit. Per-env matrices and per-suite raw G4 / G8 / G16 numbers live in `MASTER_TABLE.md` ğ1–ğ9.7, the operational source of truth.